

Configurable Voltage/Current Source Design for a Source Measure Unit (SMU)

ADI Design Challenge – Spring 2026



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Contents

Executive Summary	3
1 Introduction.....	3
1.1 About Project Report	3
1.2 Design Requirements	3
1.3 Problem Statement	4
1.4 Objective and Scope	4
2 Design Strategy	4
2.1 Why a Composite Amplifier	4
2.2 Voltage-Mode and Current-Mode Operating Principles.....	5
2.3 Stability Framework.....	5
2.4 Compensation Methods Considered	6
3 Methodology	9
3.1 Simulation Environment and Models	9
3.2 Methodology Algorithm	9
3.3 Model Validation Before Compensation	10
3.3.1 Basic Voltage-Mode and Current-Mode Verification	10
3.3.2 Operating-Point Verification	12
3.3.3 Middlebrook Loop-Gain Measurement	13
3.4 Boundary-Condition Calculations	16
Maximum Frequency for Worst-Case Capacitive Load	16
Maximum Capacitive Load at 100 kHz	17
Maximum Output Amplitude at 100 kHz with 10 μ F.....	17
Purely Resistive Load Limit	17
Discussion	17
4 Voltage-Mode Design, Stability, and Compensation	20
4.1 Baseline Configuration and Verification	20
4.2 Uncompensated Stability Across Capacitive Load.....	20
4.3 Voltage-Mode Compensation Strategy	23
4.4 Compensated Voltage-Mode Loop-Gain Results	25
4.5 Compensated Voltage-Mode Transient Results.....	25
5 Current-Mode Design, Stability, and Compensation	28
5.1 Baseline Configuration and Verification	28

5.2 Uncompensated Stability Across Capacitive Load.....	28
5.3 Current-Mode Compensation Strategy	31
5.4 Compensated Current-Mode Loop-Gain Results.....	32
6 ECSE 6050 Extension: Inductive-Load Results	34
6.1 Load Model and Rationale.....	34
6.2 Current Mode with Inductive Loads	35
6.3 Voltage Mode with Inductive Loads.....	38
7 Comparison and Design Trade-Offs	41
7.1 Performance Summary.....	41
7.2 Discussion of Trade-Offs.....	42
8 Alternative IC Options	42
9 Conclusion	43
References	44

Executive Summary

This project developed and evaluated a configurable source stage for a source measure unit that supports both voltage-output and current-output operation. A composite amplifier architecture was used, combining the ADA4661-2 as the precision stage and the AD8018 as the high-speed output stage. Stability was evaluated using the Middlebrook loop-gain method, while MATLAB was used to process LTspice data and extract frequency-domain and time-domain metrics.

The uncompensated results showed that compensation was necessary, especially under capacitive loading. Voltage mode was partially robust but still oscillatory over part of the required capacitor range, while current mode was significantly less stable. Compensation was therefore introduced using only allowed in-loop methods. The AD8018 feedback network was first constrained by the datasheet-recommended resistor values, and feedback-lead compensation provided the main stabilization effect. In current mode, the sense resistor was also reduced from $0.5\ \Omega$ to $0.05\ \Omega$ to reduce the aggressiveness of the sensing path.

After compensation, voltage mode became stable over most of the required RC load range, with only a few intermediate cases remaining marginally stable. Current mode also improved substantially, although the larger capacitive loads remained more difficult to stabilize and produced stronger transient overshoot and undershoot. For the ECSE 6050 extension, the design was further evaluated with inductive loads up to 100 mH. The compensated current-mode RL case remained stable over the full inductive range, while the voltage-mode RL case remained bounded in the time domain, though practical crossover-based margin extraction was not available. Overall, the project met the main design objectives and showed that the composite architecture is effective, with the main limitations appearing in the most demanding reactive-load cases.

1 Introduction

1.1 About Project Report

This report presents the design, simulation, and stability evaluation of a configurable source stage for a source measure unit (SMU). The work focuses directly on the implemented architecture, the compensation strategy, and the resulting loop-gain and transient behavior. Since the project was completed for the ECSE 6050 section, the analysis includes both capacitive and inductive loading cases.

1.2 Design Requirements

The design targets provided in the challenge are summarized in Table 1. The source stage must support both voltage-output and current-output operation while remaining stable over a wide range of passive loads.

Table 1: Summary of challenge specifications and design targets.

Parameter	Target
Voltage output range	-2.5 V to +2.5 V
Current output range	-200 mA to +200 mA
Maximum supply rails	± 3.5 V
Capacitive load range	400 pF to 10 μ F
Inductive load range (6050 requirement)	Up to 100 mH
Stability requirement	Stable operation in both modes across required loads
Performance emphasis	Maximize bandwidth and minimize settling time

1.3 Problem Statement

The central challenge is to realize a configurable output stage that is accurate and stable in both voltage-source and current-source operation while driving widely varying capacitive and inductive loads. The design must therefore address both static accuracy and dynamic stability, with the latter evaluated using loop-gain methods rather than relying only on closed-loop Bode plots.

1.4 Objective and Scope

The objectives of this work are:

- 1) To verify correct operation of the circuit in voltage mode and current mode.
- 2) To establish the operating boundary imposed by output-current limitations.
- 3) To evaluate uncompensated loop stability across the required capacitive-load range.
- 4) To design and justify only *allowed in-loop compensation methods*.
- 5) To verify compensated performance in both the frequency domain and time domain.
- 6) To extend the analysis to inductive loads as required for ECSE 6050.
- 7) To compare alternative IC choices relevant to the design.

2 Design Strategy

2.1 Why a Composite Amplifier

A composite amplifier structure was adopted to combine the strengths of two different amplifier stages. A precision amplifier is used to provide accurate low-frequency control, low offset, and low drift, while a high-speed output amplifier is used to provide the required drive capability and dynamic response. This arrangement is especially attractive for an SMU source stage because it allows precision and output current capability to be addressed simultaneously.

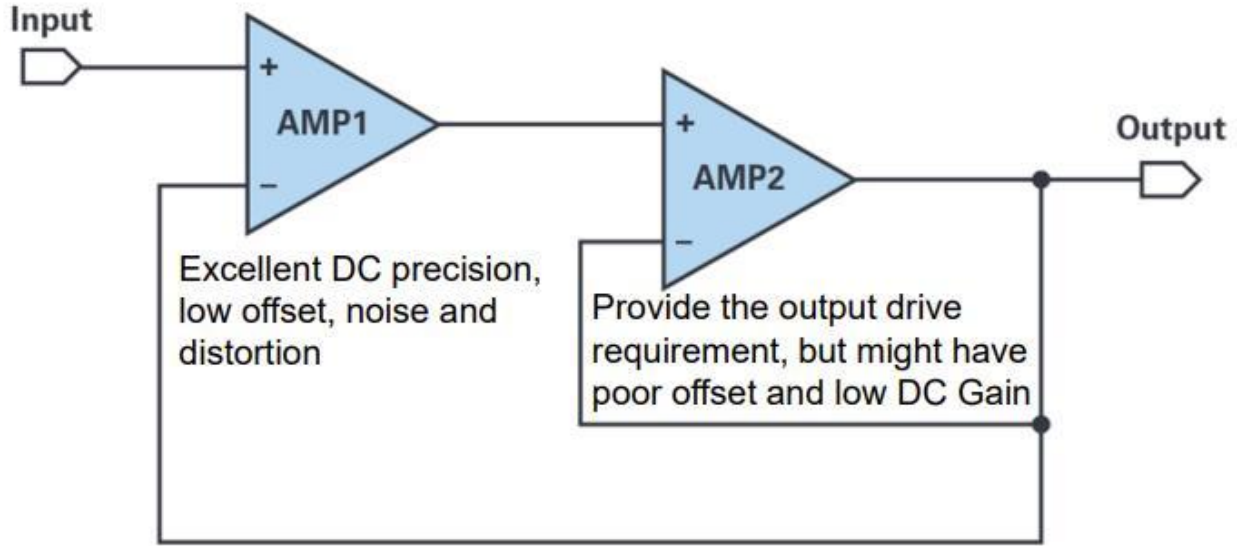


Figure 2.1: Composite amplifier architecture used to combine precision and output-drive capability.[1]

2.2 Voltage-Mode and Current-Mode Operating Principles

The circuit is designed to support two operating modes through the mode-control signal V_{mode} . In voltage mode, the feedback path is configured so that the output node tracks the commanded voltage setpoint. In current mode, the feedback path is reconfigured so that the output current is regulated through current sensing and feedback, with the current command represented as a voltage input. In both cases, the same physical output stage is retained, while V_{mode} selects the appropriate control path to realize the required operating objective.

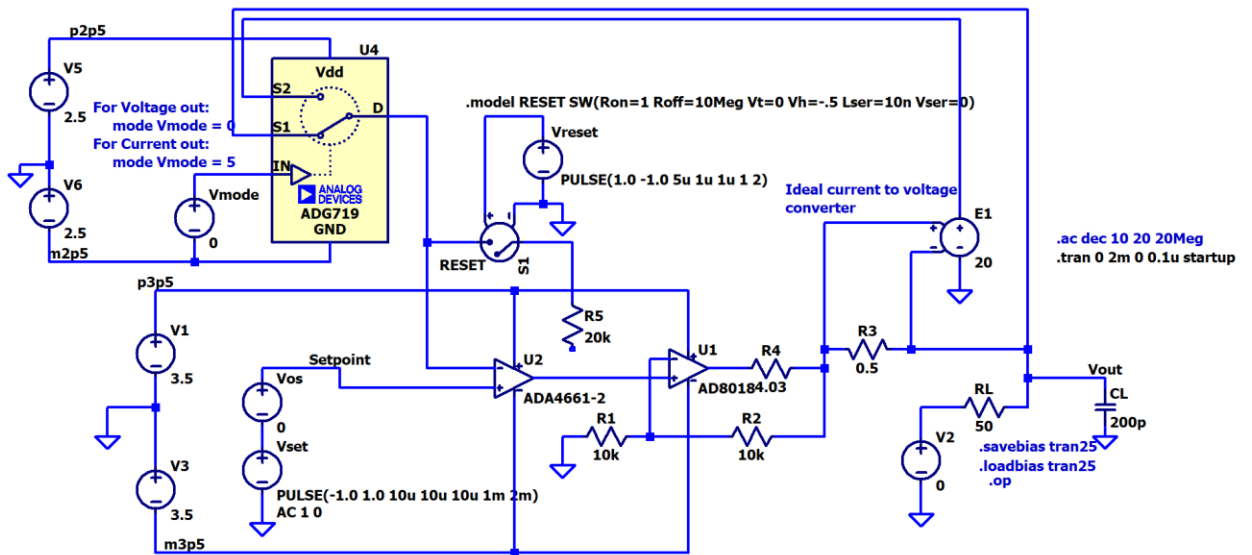


Figure 2.2: Top-level configurable SMU circuit. The operating mode is selected through the control signal, V_{mode} which reconfigures the feedback path for voltage-output or current-output operation.

2.3 Stability Framework

The stability analysis in this report is based on the loop gain $T(jf)$. The two main stability metrics

are gain margin and phase margin. Since the circuit contains multiple interacting blocks with finite input and output impedances, simple closed-loop Bode plots do not always provide enough information to judge stability. For this reason, the Middlebrook method is used to evaluate the loop gain while preserving the DC operating point.

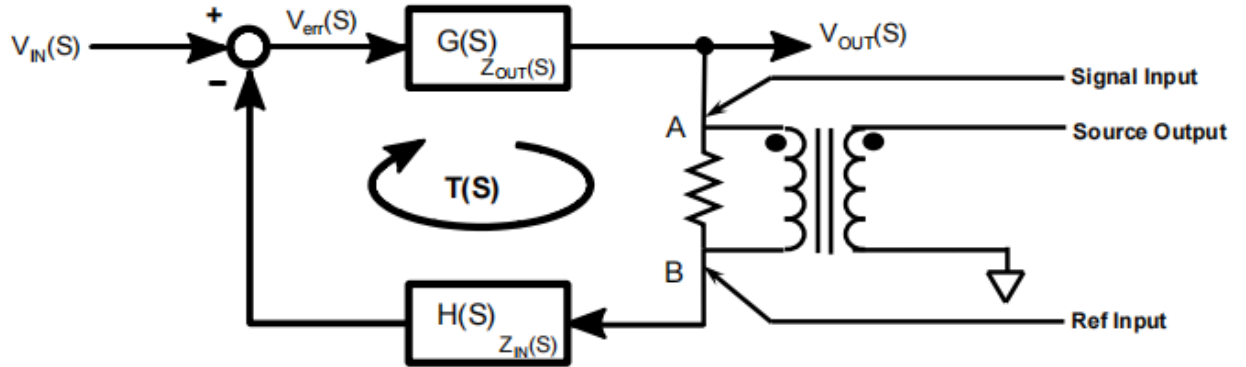


Figure 2.3: Loop-gain measurement setup using the Middlebrook injection method. [1]

2.4 Compensation Methods Considered

Only compensation approaches that remain inside the intended control loop are considered as final design solutions in this report. Based on the course guidance, external output snubber compensation and the prohibited output-side feedback-capacitor method are not used as final project solutions. The compensation methods considered are summarized in Table 2.

Table 2: Compensation methods considered for the project, indicating which are permitted, which were rejected, and why.

Method	Status	Reason
Input lag compensation	Used if needed	In-loop method; shapes loop crossover and improves phase margin
Sensor-loop compensation	Used if needed	In-loop method; stabilizes current-sense/transimpedance path
Datasheet-constrained feedback resistor selection	Used	Sets allowable bandwidth and stability region of output amplifier
Output snubber network	Not Used	Not used in the final design because it stabilizes the circuit by modifying the external output loading rather than by shaping the intended in-loop feedback path.
Output-side feedback capacitor method	Not Used	Not used in the final design because it is outside the intended in-loop compensation path and is not permitted by the project guidance.
Feedback lead compensation	Used	In-loop method; introduces a zero-pole shaping effect in the feedback path to improve phase margin and reduce the rate of closure at crossover.

Input-Lag Compensation

The input-lag compensation network used in this work is shown in Figure 1. The network consists of a series R_c - C_c branch connected between the inverting and non-inverting input nodes of the composite amplifier stage. Its purpose is to reshape the feedback factor so that the loop gain experiences improved phase margin near the crossover frequency.

Assuming an ideal amplifier block a , the compensation network modifies the feedback response such that

$$H(s) \approx \frac{1+sR_cC_c}{1+s(R_c+R_1\parallel R_2)C_c}. \quad (1)$$

This introduces a zero at

$$f_z = \frac{1}{2\pi R_c C_c}, \quad (2)$$

and a pole at

$$f_p = \frac{1}{2\pi(R_c+R_1\parallel R_2)C_c}. \quad (3)$$

Since

$$R_c + R_1 \parallel R_2 > R_c, \quad (4)$$

it follows that

$$f_z < f_p. \quad (5)$$

Therefore, the inverse feedback factor $1/\beta$ remains approximately constant at low frequency, with a low-frequency value given by

$$\left. \frac{1}{\beta} \right|_{f \rightarrow 0} \approx 1 + \frac{R_2}{R_1}, \quad (6)$$

then begins to rise with a slope of approximately +20 dB/dec after f_z , and finally flattens again after f_p . This behavior is illustrated in Figure 1(b).

The practical effect of this network is to create a phase-lead region between f_z and f_p . By properly placing this region near the loop crossover frequency, the phase margin can be increased without excessively reducing the closed-loop bandwidth. In the present design, this compensation method is especially useful because it remains inside the control loop and improves stability through feedback shaping rather than through an external output-side loading method.

For effective compensation, the zero is placed below the intended crossover frequency so that the phase boost is already active as the loop approaches unity gain. The pole is then placed at a higher frequency to limit the high-frequency effect of the compensation branch. In this way, the network improves damping

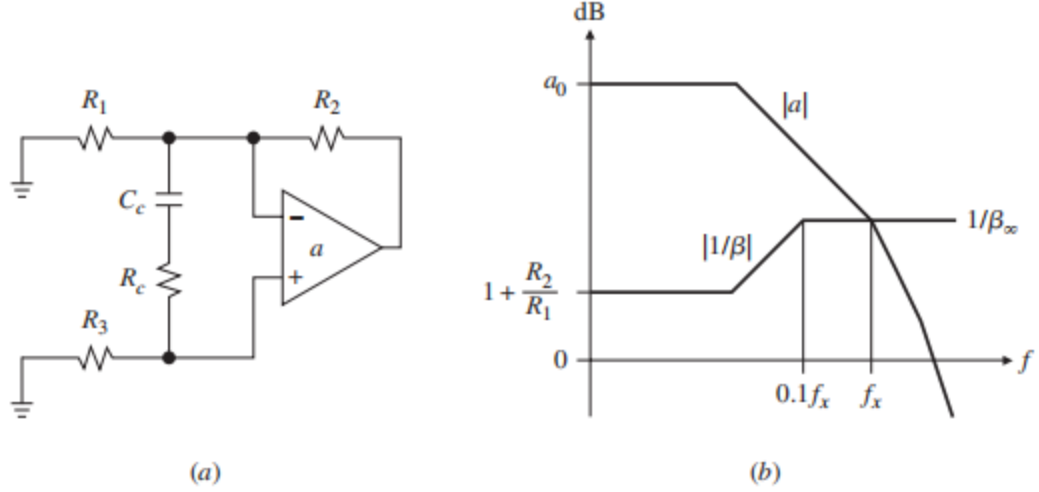


Figure 2.4: Input-lag compensation network and its effect on the inverse feedback factor $1/|\beta|$. [3]

Datasheet Constrained Feedback Resistor Selection

The AD8018 is a current-feedback amplifier, so its feedback resistor cannot be selected solely from the desired closed-loop gain. In current-feedback amplifiers, the feedback resistor directly influences bandwidth, gain flatness, and stability. The AD8018 datasheet states that the amplifier bandwidth depends primarily on the feedback resistor and that there is a value of R_F below which the amplifier may become unstable due to additional poles. For this reason, the manufacturer's recommended resistor values were used as the initial design point. For a gain of +2, the datasheet recommends $R_F = 750$ and $R_G = 750$. These values were therefore adopted as the starting point for the output stage, and all subsequent compensation design was performed around this datasheet-constrained choice. This approach avoids artificially increasing the loop crossover frequency by reducing R_F into a range that could degrade phase margin or cause instability.

Feedback Lead Compensation

The feedback lead compensator places a capacitor C_f across the feedback resistor R_2 . At low frequency, C_f is effectively open, so the nominal low-frequency gain remains unchanged. At higher frequency, the impedance of C_f decreases, which reshapes $1/\beta$, reduces the rate of closure near crossover, and improves phase margin. A commonly used design expression for the feedback capacitor is

$$C_f = \frac{\sqrt{1 + \frac{R_2}{R_1}}}{2\pi R_2 f_x},$$

where f_x is the intended crossover frequency.

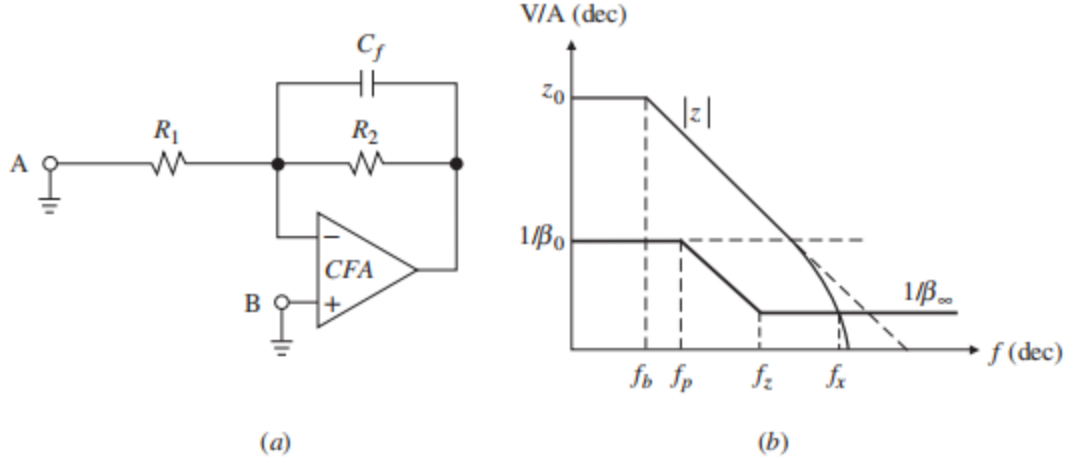


Figure 2.5: Feedback Lead Compensation

3 Methodology

3.1 Simulation Environment and Models

All circuit simulations were performed in LTspice using test-bench-1. MATLAB was used to post-process loop-gain data exported from LTspice, generate cleaned Bode plots, and extract gain crossover frequency, phase crossover frequency, phase margin, and gain margin. Where necessary, convergence support using .op, .savebias, .loadbias, startup directives, and reset-assisted initialization was applied.

3.2 Methodology Algorithm

1. Initial circuit verification

The circuit was first configured in both voltage-output and current-output modes to verify that the mode-selection network, amplifier connections, and sensing paths operated as intended.

2. Model validation before compensation

Basic .op, transient, and AC simulations were performed to confirm correct DC biasing, correct output tracking, and proper signal flow in both voltage mode and current mode.

3. Loop identification and Middlebrook implementation

The feedback loop was identified for each operating mode, and a suitable injection point was selected. Voltage injection and current injection were then implemented to obtain partial loop-gain measurements, which were combined to extract the full loop gain without disturbing the DC operating point.

4. Boundary-condition analysis

Hand calculations were used to determine the feasible combinations of output amplitude, frequency, and load capacitance under the output-current limit of the selected output-stage amplifier.

5. Initial sweep for instability

The uncompensated circuit was swept across the required capacitive-load range to identify unstable and marginally stable regions in both voltage mode and current mode.

6. Apply allowed in-loop compensation techniques

Compensation methods that remain inside the intended control loop were selected and applied. Methods not in the feedback loop were not used as final design solutions.

7. Validation through resweep

After compensation, loop-gain and transient analyses were repeated across the required load ranges to confirm improved stability and acceptable dynamic response.

8. Extension to inductive loads

Since this work was completed for ECSE 6050, the same process was extended to inductive-load operation.

9. Iteration if needed

If the design failed to meet the desired stability or performance targets, the compensation values were revised and the analysis was repeated.

Figure 5: Methodology algorithm used to verify, stabilize, and validate the configurable SMU output stage.

3.3 Model Validation Before Compensation

3.3.1 Basic Voltage-Mode and Current-Mode Verification

Before beginning the compensation design, the baseline test bench (Figure 2.2) was verified in both voltage-output and current-output configurations. This stage follows the same purpose as Checkpoint 1, where the immediate goal was to confirm that the circuit could be configured correctly and that the expected input-output relationships were visible in simulation.

In voltage-output mode, transient analysis was used to demonstrate correct tracking of the input setpoint by the output voltage. In current-output mode, transient analysis was used to demonstrate correct regulation of the output current relative to the input setpoint. Similarly, AC analysis was performed in both modes to confirm the corresponding small-signal relationships.

Voltage-mode simulation converged normally without additional initialization. For current mode, convergence was significantly slower, so the initial condition of the output capacitor was set to 0 V to assist startup. This provided a stable initialization point and allowed the current-mode simulations to run reliably.

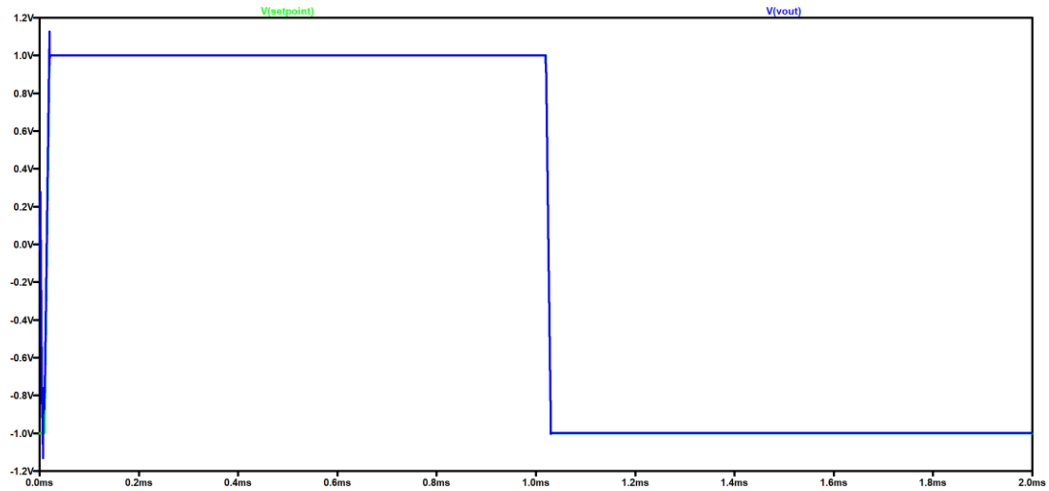


Figure 3.1: Pre-compensation voltage-mode transient verification showing correct output tracking.

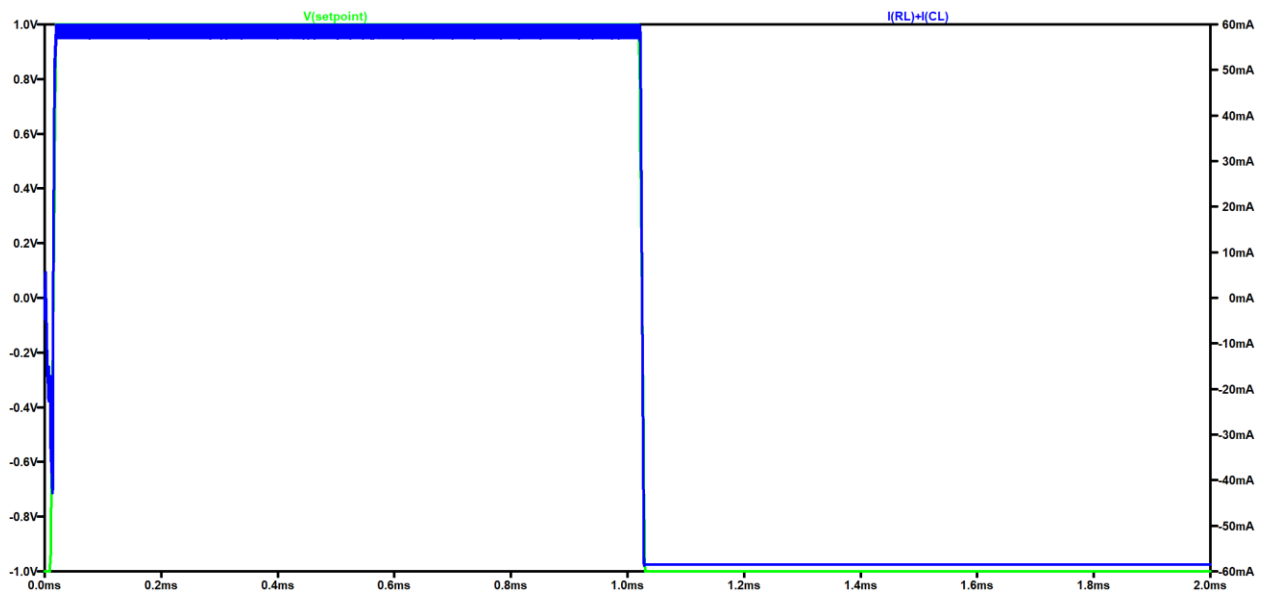


Figure 3.2: Pre-compensation current-mode transient verification showing correct input-to-output-current behavior.

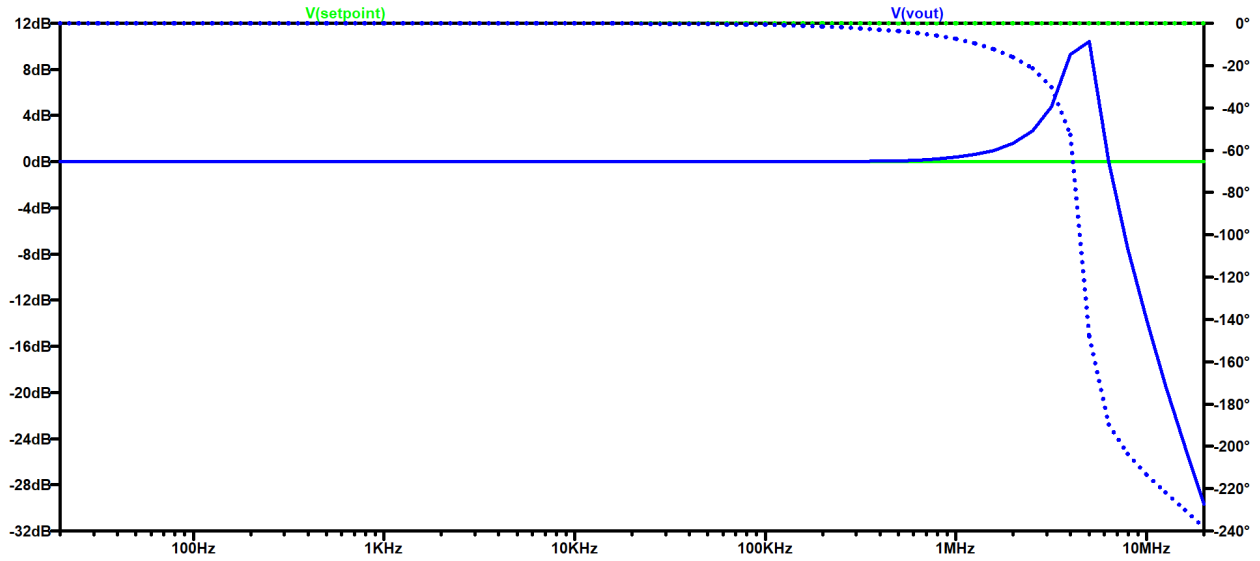


Figure 3.3: Pre-compensation AC verification in voltage mode relating input and output voltage.

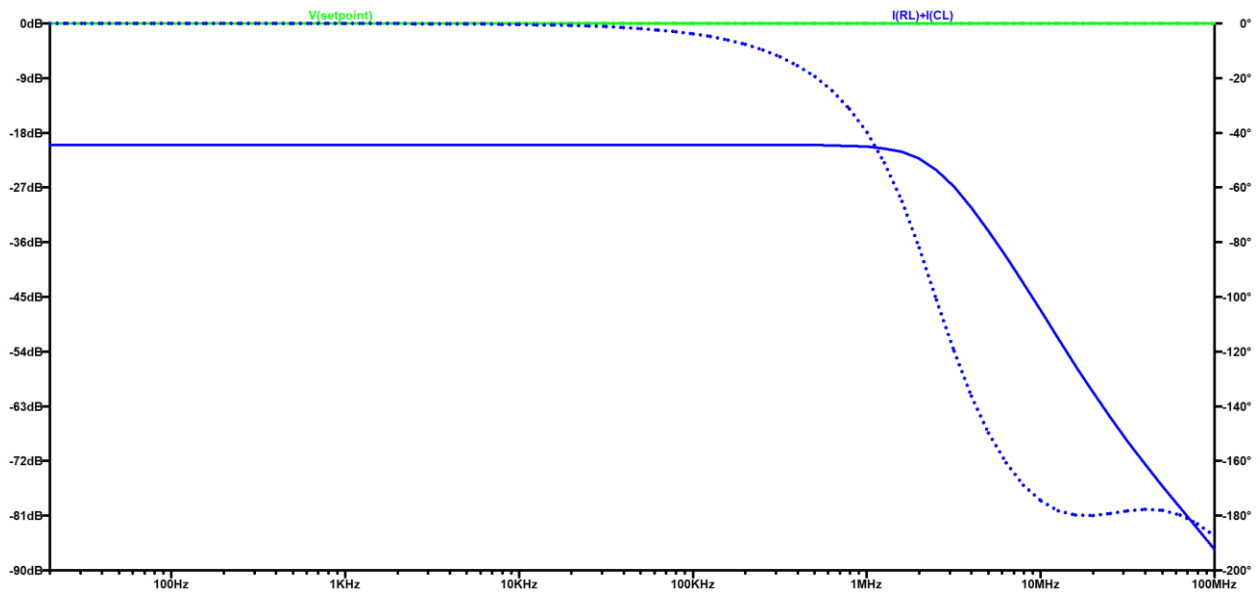


Figure 3.4: Pre-compensation AC verification in current mode relating input and output current.

3.3.2 Operating-Point Verification

An operating-point analysis was then performed to confirm valid DC bias conditions. The .op results were used to verify that all key node voltages remained within the available supply rails, that the output reflected the intended mode configuration, and that the simulation converged properly before any frequency-domain or transient stability study was attempted.

Table 3:: Key operating-point quantities confirming valid DC bias conditions prior to compensation design.

Node / Quantity	Value	Interpretation
$V(\text{set})$	[-1 V]	Input setpoint
$V(\text{out})$	[-0.9995 V]	Output operating point
$V(+3.5\text{rail})$	[+3.5 V]	Positive supply rail
$V(-3.5\text{rail})$	[-3.5 V]	Negative supply rail
$I(R_L)$	[-0.0199906 V]	Resistive-load current
$I(C_L)$	[0]	Capacitor DC current, expected near zero

3.3.3 Middlebrook Loop-Gain Measurement

After the basic circuit operation was verified, the Middlebrook loop-gain procedure was implemented in both voltage mode and current mode. This stage follows the same purpose as Checkpoint 2, where the objective was to identify the feedback loop, select a suitable injection point, and obtain the loop gain using voltage injection and current injection.

The loop was broken at a point chosen to remain common to the relevant control path while preserving the original DC operating point. A floating AC voltage source was first inserted at the break location to perform voltage injection. The forward and return voltages adjacent to the source were measured, and the partial loop gain was computed as

$$T_V(jf) = -\frac{V_r}{V_f}. \quad (7)$$

A current source was then inserted in parallel at the same loop-break location to perform current injection. The forward and return currents were measured, and the second partial loop gain was computed as

$$T_I(jf) = -\frac{i_r}{i_f}. \quad (8)$$

The full loop gain was assembled from the two measurements using

$$\frac{1}{1+T} = \frac{1}{1+T_V} + \frac{1}{1+T_I}, \quad (9)$$

which gives

$$T = \frac{T_V T_I - 1}{T_V + T_I + 2}. \quad (10)$$

A common loop-break location was selected for both voltage-output and current-output operation so that the partial loop gains obtained from voltage injection and current injection could be evaluated consistently in both modes. This provided a uniform Middlebrook measurement procedure while preserving the DC operating point.

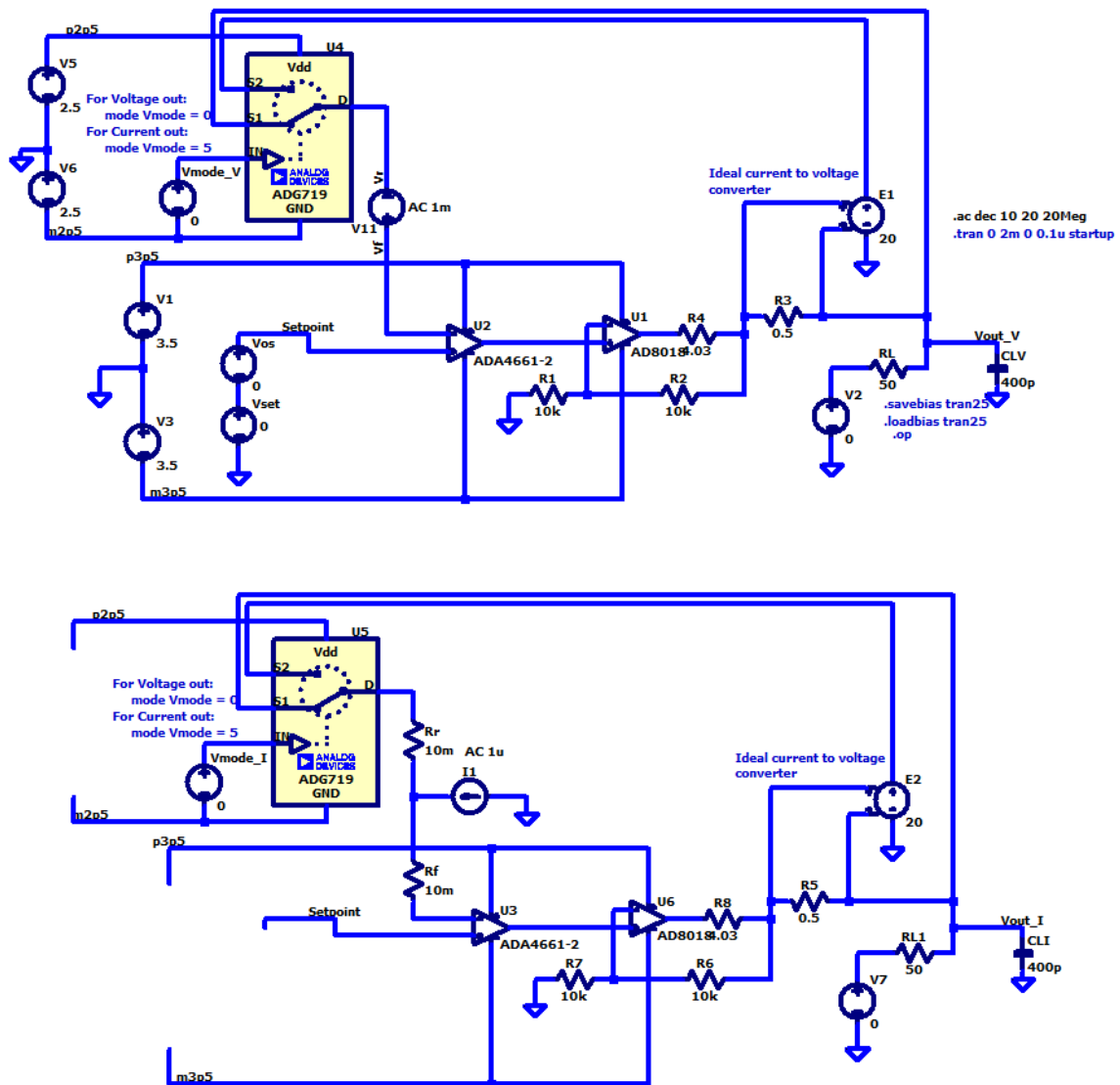


Figure 3.5: Common loop-break location used for Middlebrook loop-gain extraction in both voltage mode and current mode.

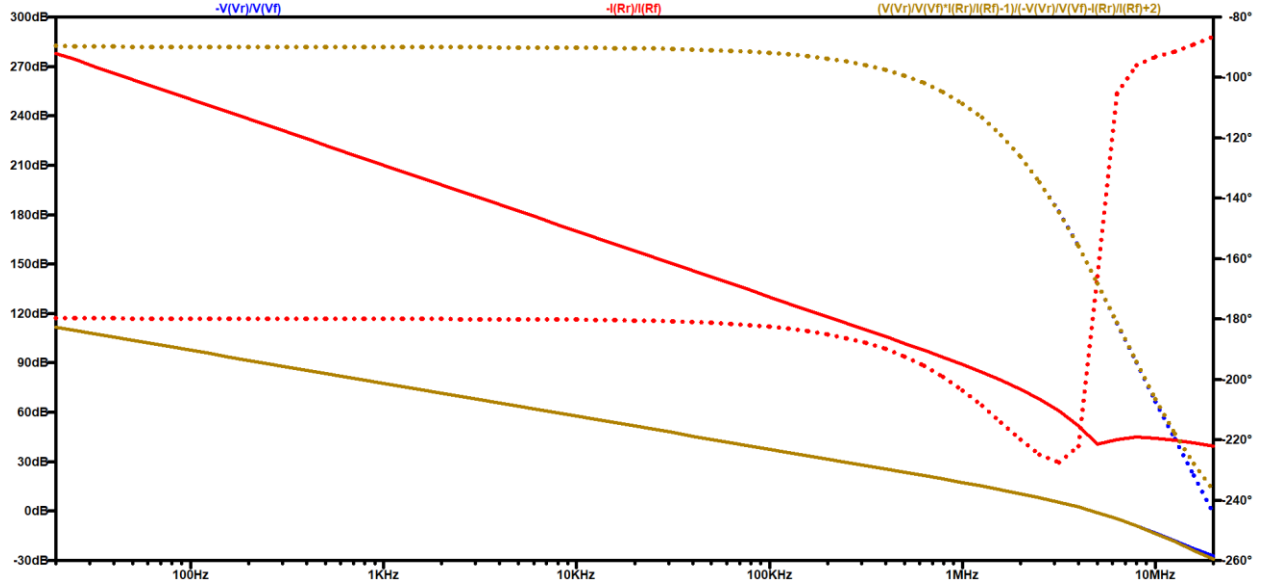


Figure 3.6: Voltage-mode comparison of T_V , T_I , and assembled loop gain T , showing that T_V closely approximates T .

The assembled loop gain T in voltage mode is seen to track T_V almost exactly, while T_I remains substantially larger and does not govern the crossover region. Accordingly, T_V was taken as the simplified approximation to the full loop gain for voltage-mode analysis from combined injection.

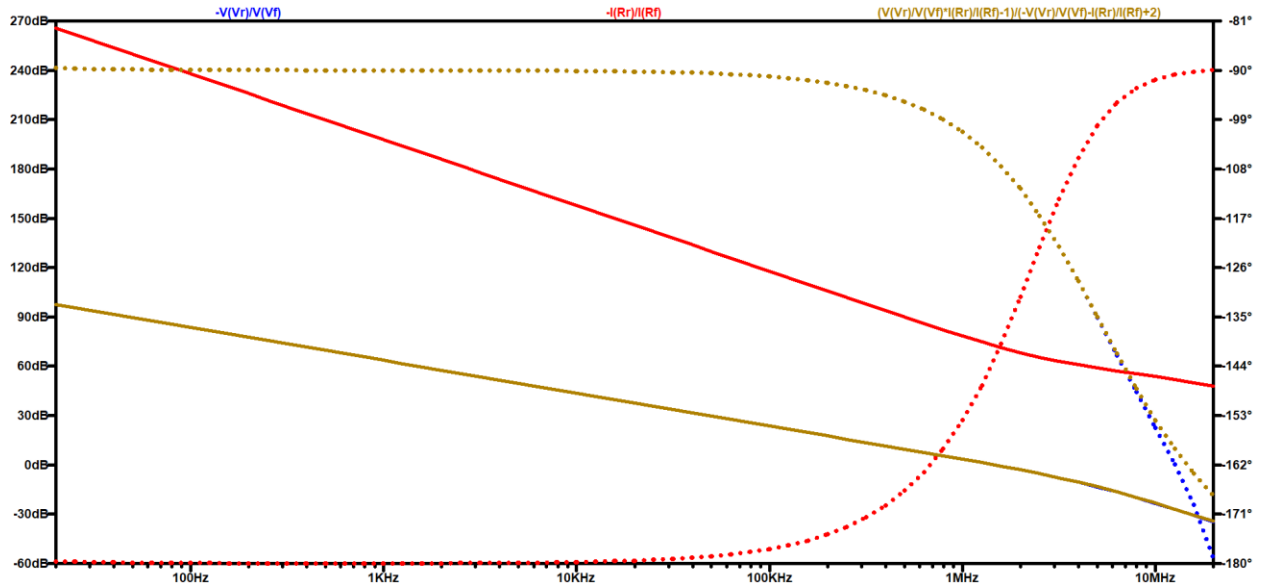


Figure 3.7: Current-mode comparison of T_V , T_I , and assembled loop gain T , showing that T_V provides the closer approximation to T .

In current mode, although both partial loop gains were evaluated, the assembled loop gain T remains much closer to the voltage-injection result T_V than to the current-injection result T_I , particularly in the crossover region. Accordingly, T_V was used as the simplified approximation to the full loop gain for current-mode analysis.

In both voltage mode and current mode, the assembled loop gain T closely overlaps with the voltage-injection result T_V , particularly in the crossover region. Therefore, T_V was used as the simplified approximation to the full loop gain for subsequent stability analysis.

3.4 Boundary-Condition Calculations

The output-stage boundary conditions were evaluated using the peak-current capability of the AD8018 together with the specified maximum output amplitude, maximum frequency, and required capacitive-load range. The challenge defines the relevant worst-case design quantities as a maximum output amplitude of ± 2.5 V, a maximum sinewave frequency of 100 kHz, and capacitive loads up to 10 μ F. The AD8018 datasheet is consistent with this framework and lists linear output current in the range of approximately 350mA to 400mA under the stated conditions.

For a sinusoidal output voltage

$$v_o(t) = V_p \sin(2\pi f t), \quad (11)$$

the capacitor current is

$$i_C(t) = C_L \frac{dv_o(t)}{dt}. \quad (12)$$

Therefore, the peak capacitor current is

$$I_{C,pk} = 2\pi f C_L V_p. \quad (13)$$

If a resistive load R_L is also present in parallel, then the resistor draws a peak current of

$$I_{R,pk} = \frac{V_p}{R_L}. \quad (14)$$

Hence, the total peak output-current requirement in voltage mode becomes

$$I_{out,pk} = \frac{V_p}{R_L} + 2\pi f C_L V_p. \quad (15)$$

Maximum Frequency for Worst-Case Capacitive Load

For the ideal capacitive-load case, the maximum allowable operating frequency is obtained from

$$f_{max} = \frac{I_{out,pk,max}}{2\pi C_L V_p}. \quad (16)$$

Using

$$I_{out,pk,max} = 0.4 \text{ A}, \quad C_L = 10 \mu\text{F}, \quad V_p = 2.5 \text{ V},$$

gives

$$f_{max} = \frac{0.4}{2\pi(10 \times 10^{-6})(2.5)} \approx 2.55 \text{ kHz}. \quad (17)$$

If a 50 Ω resistive load is present in parallel, the resistor alone draws

$$I_{R,pk} = \frac{2.5}{50} = 0.05 \text{ A} = 50 \text{ mA}. \quad (18)$$

This leaves only

$$I_{C,pk,max} = 0.4 - 0.05 = 0.35 \text{ A} \quad (19)$$

available for charging the capacitor. The corresponding maximum frequency is therefore

$$f_{max} = \frac{0.35}{2\pi(10 \times 10^{-6})(2.5)} \approx 2.23 \text{ kHz}. \quad (20)$$

Maximum Capacitive Load at 100 kHz

For a fixed operating frequency of 100 kHz, the maximum allowable capacitance is

$$C_{L,\max} = \frac{I_{\text{out,pk,max}}}{2\pi f V_p}. \quad (21)$$

Using

$$I_{\text{out,pk,max}} = 0.4 \text{ A}, \quad f = 100 \text{ kHz}, \quad V_p = 2.5 \text{ V},$$

gives

$$C_{L,\max} = \frac{0.4}{2\pi(100 \times 10^3)(2.5)} \approx 0.255 \text{ } \mu\text{F}. \quad (22)$$

If a 50 Ω resistive load is also present, only 0.35 A remains available for the capacitor, so

$$C_{L,\max} = \frac{0.35}{2\pi(100 \times 10^3)(2.5)} \approx 0.223 \text{ } \mu\text{F}. \quad (23)$$

Maximum Output Amplitude at 100 kHz with 10 μF

The maximum allowable output amplitude for a fixed frequency and capacitance may also be written as

$$V_{p,\max} = \frac{I_{\text{out,pk,max}}}{\frac{1}{R_L} + 2\pi f C_L}. \quad (24)$$

For

$$I_{\text{out,pk,max}} = 0.4 \text{ A}, \quad R_L = 50 \text{ } \Omega, \quad f = 100 \text{ kHz}, \quad C_L = 10 \text{ } \mu\text{F},$$

the result is

$$V_{p,\max} = \frac{0.4}{0.02 + 2\pi(100 \times 10^3)(10 \times 10^{-6})} \approx 63.5 \text{ mV}. \quad (25)$$

Purely Resistive Load Limit

For a purely resistive load, the minimum allowable load resistance based on the AD8018 peak-current capability is

$$R_{L,\min} = \frac{V_p}{I_{\text{out,pk,max}}}. \quad (26)$$

Using $V_p = 2.5 \text{ V}$ and $I_{\text{out,pk,max}} = 0.4 \text{ A}$ gives

$$R_{L,\min} = \frac{2.5}{0.4} = 6.25 \text{ } \Omega. \quad (27)$$

If the stricter SMU operating specification of $\pm 200 \text{ mA}$ is enforced instead, then the corresponding minimum resistive load is

$$R_{L,\min} = \frac{2.5}{0.2} = 12.5 \text{ } \Omega. \quad (28)$$

Discussion

These calculations show that the full $\pm 2.5 \text{ V}$ output swing cannot be maintained simultaneously

with very high frequency and very large capacitance. In particular, a $10\ \mu\text{F}$ load can only be driven to the full amplitude at a few kilohertz, while operation at $100\ \text{kHz}$ requires the maximum capacitive load to be reduced to approximately $0.255\ \mu\text{F}$, or about $0.223\ \mu\text{F}$ when a $50\ \Omega$ resistive load is also present. The boundary-condition analysis therefore defines the feasible operating surface of the source stage and provides a quantitative basis for the subsequent stability study.

Table 4: Current-limit trade-offs among output amplitude, frequency, and capacitance.

Scenario	Given	Result
High-speed operation, no resistive load	$V_p = 2.5\ \text{V}, f = 100\ \text{kHz}$	$C_{L,\text{max}} \approx 0.255\ \mu\text{F}$
High-speed operation with $50\ \Omega$ load	$V_p = 2.5\ \text{V}, f = 100\ \text{kHz}, R_L = 50\ \Omega$	$C_{L,\text{max}} \approx 0.223\ \mu\text{F}$
Worst-case large capacitance, no resistive load	$V_p = 2.5\ \text{V}, C_L = 10\ \mu\text{F}$	$f_{\text{max}} \approx 2.55\ \text{kHz}$
Worst-case large capacitance with $50\ \Omega$ load	$V_p = 2.5\ \text{V}, C_L = 10\ \mu\text{F}, R_L = 50\ \Omega$	$f_{\text{max}} \approx 2.23\ \text{kHz}$
Maximum voltage at $100\ \text{kHz}$ and $10\ \mu\text{F}$	$f = 100\ \text{kHz}, C_L = 10\ \mu\text{F}, R_L = 50\ \Omega$	$V_{p,\text{max}} \approx 63.5\ \text{mV}$
Purely resistive load limit from AD8018 peak-current capability	$V_p = 2.5\ \text{V}, I_{\text{pk,max}} = 400\ \text{mA}$	$R_{L,\text{min}} = 6.25\ \Omega$
Purely resistive load limit from SMU output-current specification	$V_p = 2.5\ \text{V}, I_{\text{pk,max}} = 200\ \text{mA}$	$R_{L,\text{min}} = 12.5\ \Omega$

The MATLAB boundary plots confirm the analytical current-limit trade-off among frequency, output amplitude, and capacitive load. For the ideal capacitive-load case, the maximum capacitance that can be driven at $100\ \text{kHz}$ and $V_p = 2.5\ \text{V}$ is approximately $0.255\ \mu\text{F}$. When a $50\ \Omega$ resistive load is also present, the available capacitor-charging current is reduced, and the maximum allowable capacitance at the same operating point drops to approximately $0.223\ \mu\text{F}$.

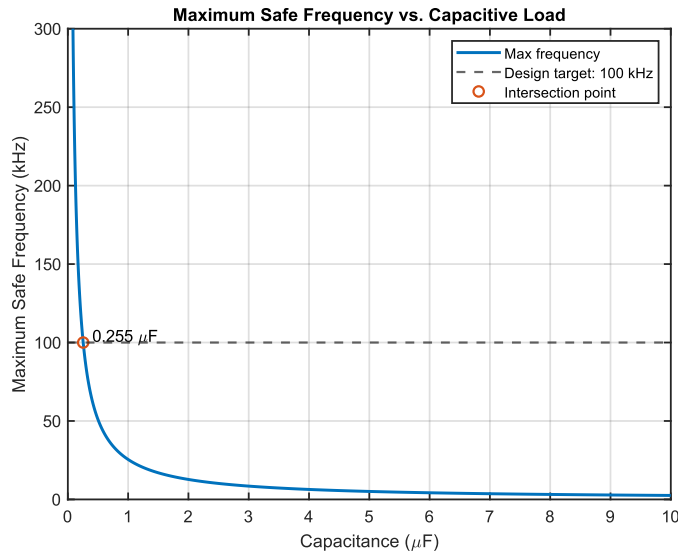


Figure 3.8: Maximum safe frequency as a function of capacitive load based on the 400mA peak-current limit of the AD8018 for the ideal capacitive-load case.

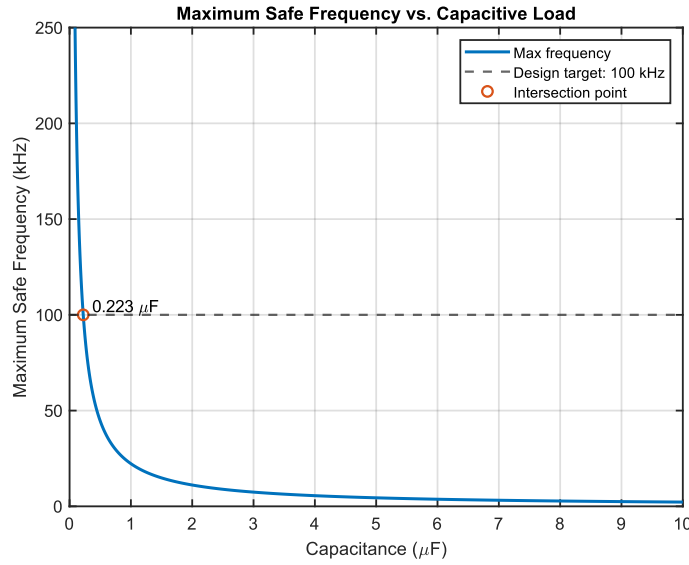


Figure 3.9: Maximum safe frequency as a function of capacitive load when a $50\ \Omega$ resistive load is present in parallel with the capacitor, reducing the current available for capacitive drive.

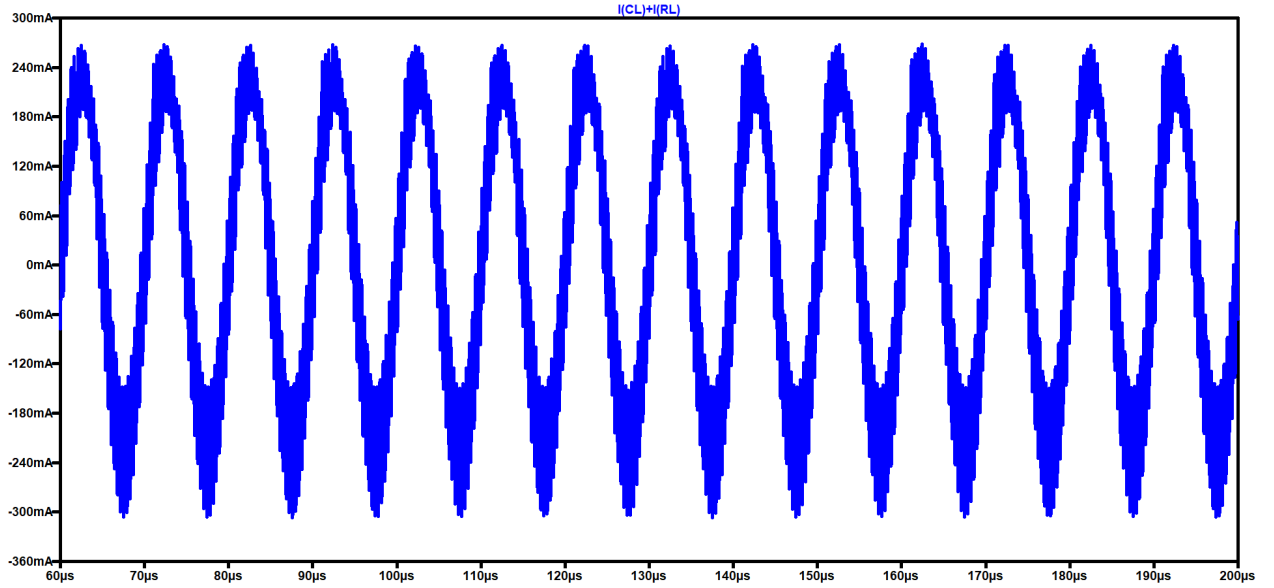


Figure 43.10: Simulated total output current $I(C_L) + I(R_L)$ at the analytical RC boundary condition for $V_{set} = 2.5\text{ V}$ amplitude, $f = 100\text{ kHz}$, $R_L = 50\ \Omega$, and $C_L = 0.223\ \mu\text{F}$. The uncompensated circuit shows strong oscillatory peaking, so the instantaneous current departs from the ideal sinusoidal boundary estimate

Figure 3.10 shows the simulated total output current, $I(C_L) + I(R_L)$, at the analytically predicted RC boundary condition for $V_{set} = 2.5\text{ V}$ amplitude, $f = 100\text{ kHz}$, $R_L = 50\ \Omega$, and $C_L = 0.223\ \mu\text{F}$. From the hand calculation, this operating point corresponds to the nominal current-limit boundary for the selected output stage under ideal sinusoidal behavior. However, the uncompensated transient response does not remain purely sinusoidal. Instead, the waveform becomes highly oscillatory, and the instantaneous current exhibits significant peaking. This indicates that the analytical boundary defines only the ideal current-limit condition, whereas the

actual uncompensated circuit develops instability-induced distortion and current peaking near this operating point. The result therefore reinforces the need for a stability analysis in addition to the boundary-condition calculation.

4 Voltage-Mode Design, Stability, and Compensation

4.1 Baseline Configuration and Verification

This section presents the baseline uncompensated voltage-mode configuration used as the starting point for the stability study. The circuit was first verified in transient simulation to confirm correct voltage-output operation before any compensation was introduced. In the AD8018 output stage, the feedback resistor was set to the datasheet-recommended starting value for gain +2, so that the uncompensated baseline remained consistent with current-feedback amplifier stability guidance. To support the later capacitive-load study, the output capacitor was parameterized as C_L , allowing either a single-value run or a stepped sweep across the required load range.

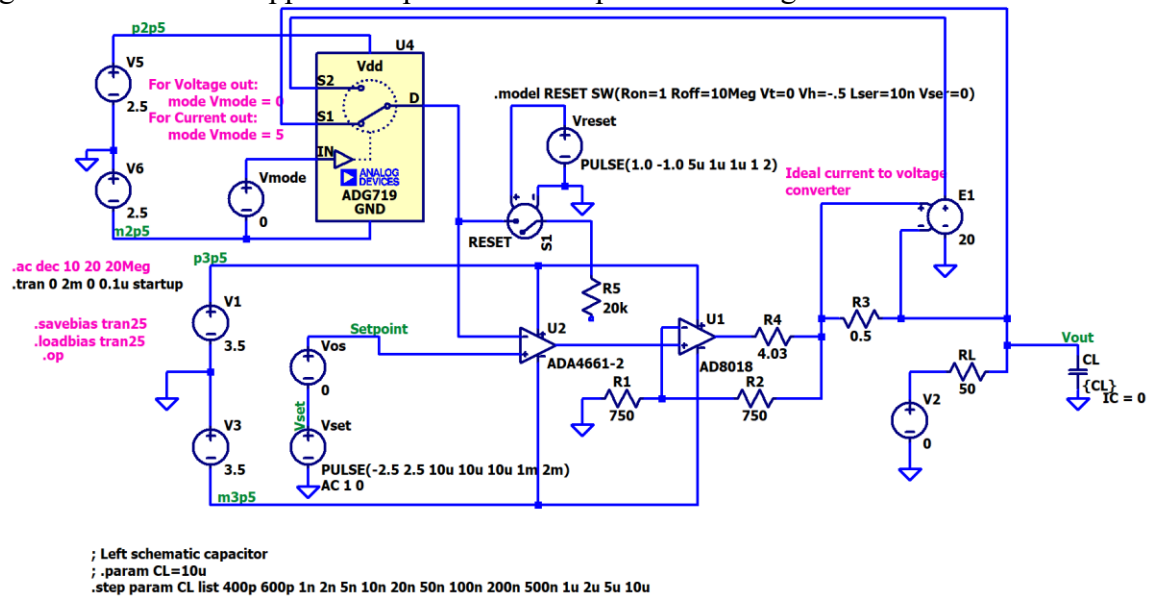


Figure 4.1 Baseline uncompensated circuit configured in voltage-output mode for initial transient verification. The AD8018 feedback network uses the datasheet-recommended starting resistor values for gain +2, and the output capacitor is parameterized as C_L to allow either a single selected value or a capacitive-load sweep.

4.2 Uncompensated Stability Across Capacitive Load

The uncompensated voltage-mode circuit exhibits oscillatory transient behavior across the capacitive-load sweep, as shown in Figure 4.2, indicating that the baseline loop is not adequately damped. The corresponding loop-gain behavior is shown in Figure 4.3 from LTspice and in Figure 4.4 after MATLAB post-processing of the exported data. The extracted stability metrics in Figure 4.5, also obtained from the MATLAB-processed data, confirm that the uncompensated design is not robust over the full required load range. The loop is stable only for the smallest and largest

capacitive loads, specifically 400 pF, 1 μ F, 2 μ F, 5 μ F, and 10 μ F. Most intermediate cases are only marginally stable, while the 50 nF case is unstable, with both negative phase margin and negative gain margin. These results show that compensation is required before the voltage-mode design can satisfy the full capacitive-load specification.

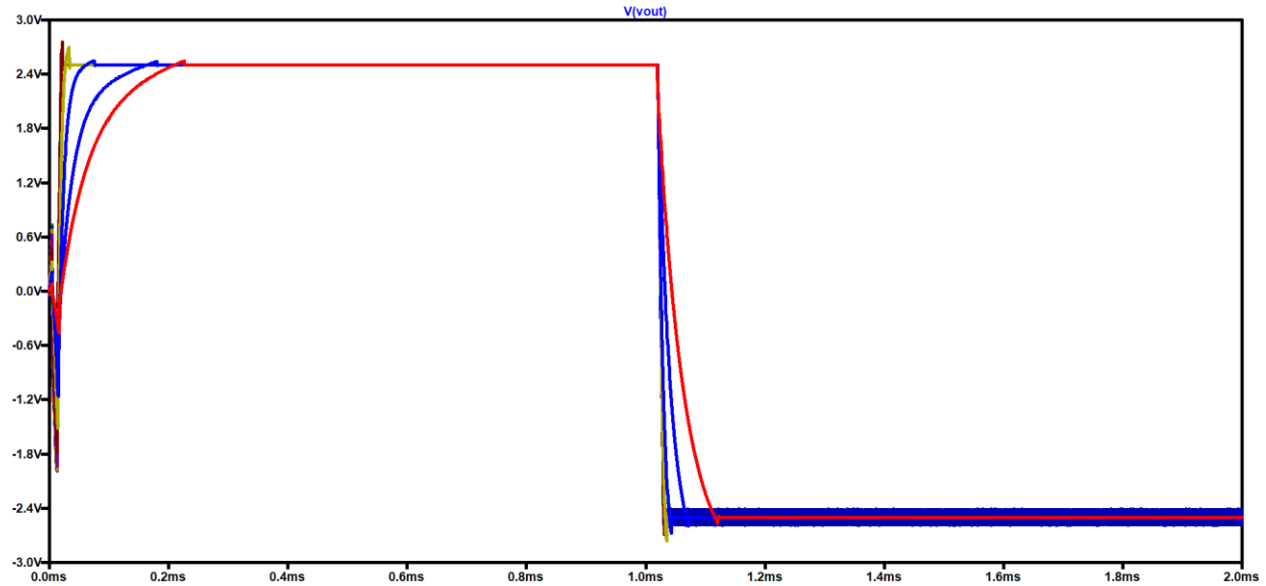


Figure 4.2: Uncompensated voltage-mode transient response across the capacitive-load sweep, showing oscillatory behavior and limited damping.

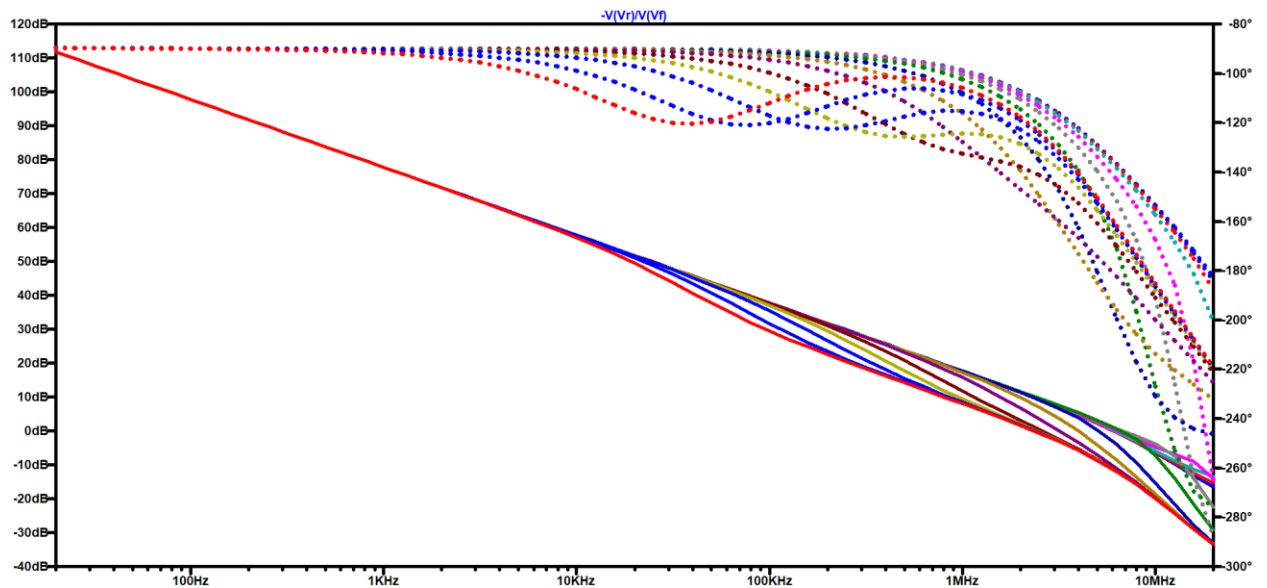


Figure 4.3: LTspice uncompensated voltage-mode loop-gain plot obtained using the Middlebrook voltage-injection method across the capacitive-load sweep

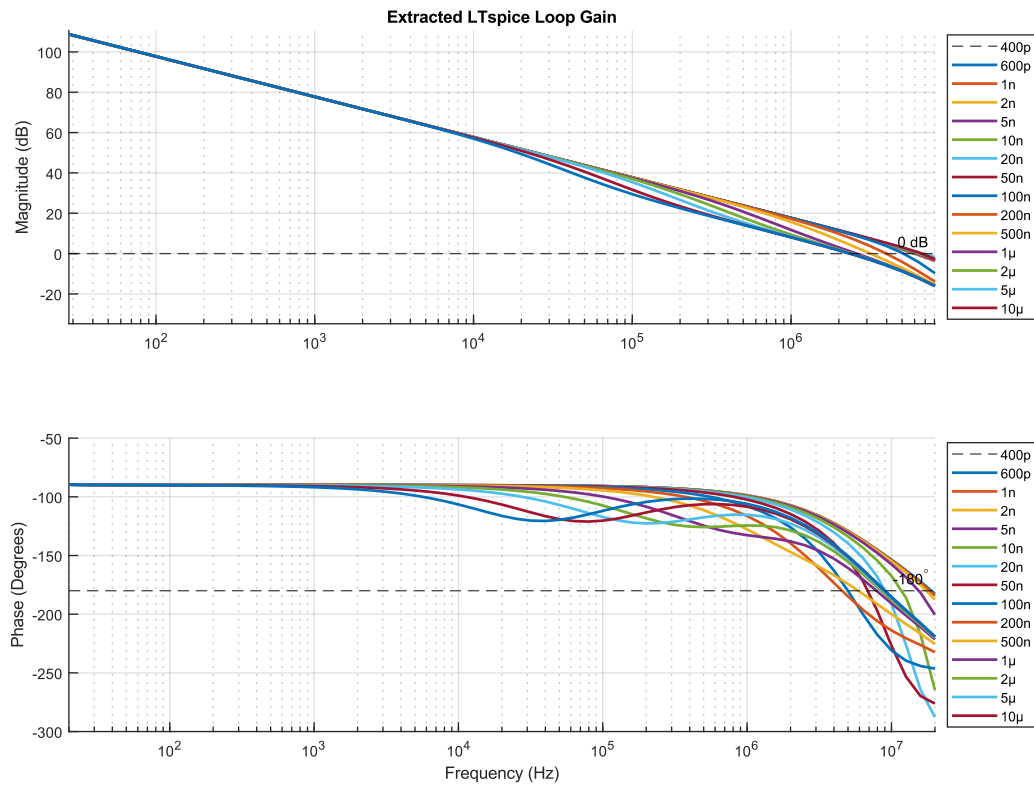


Figure 4.4: MATLAB-processed Bode plot of the uncompensated voltage-mode loop gain generated from the exported LTspice data.

	CL	f _{gc} (MHz)	f _{pc} (MHz)	GM (dB)	PM (deg)	Status
1	400p	6.0020	18.8839	15.6690	45.0140	Stable
2	600p	6.0176	18.3704	14.9610	44.6910	Marginally Stable
3	1n	6.0493	17.3987	13.6440	44.0330	Marginally Stable
4	2n	6.1301	15.4366	11.0420	42.3100	Marginally Stable
5	5n	6.3806	11.7038	6.2270	36.3950	Marginally Stable
6	10n	6.7632	8.9815	2.8030	23.4720	Marginally Stable
7	20n	6.6301	6.8319	0.4090	3.2920	Marginally Stable
8	50n	5.0989	4.9643	-0.4010	-1.8480	Unstable
9	100n	3.9788	4.5502	2.4000	6.8940	Marginally Stable
10	200n	3.1675	5.8760	9.8500	20.1650	Marginally Stable
11	500n	2.5853	7.8670	15.5600	38.1650	Marginally Stable
12	1μ	2.4222	8.4970	16.9870	47.8960	Stable
13	2μ	2.3686	8.7957	17.6100	53.4100	Stable
14	5μ	2.3508	8.9692	17.9550	56.7830	Stable
15	10μ	2.3476	9.0256	18.0650	57.8930	Stable

Figure 4.5: Stability metrics extracted in MATLAB from the exported uncompensated voltage-mode loop-gain data, including gain crossover frequency, phase crossover frequency, gain margin, phase margin, and stability classification across the capacitive-load sweep.

4.3 Voltage-Mode Compensation Strategy

The uncompensated voltage-mode results in Section 4.2 showed that the loop is not robust over the required capacitive-load range, with several intermediate load cases exhibiting only marginal stability and the 50 nF case becoming unstable. Compensation was therefore introduced in voltage mode using three elements: a datasheet-constrained feedback network for the AD8018 output stage, a feedback-lead capacitor, and an input-lag network.

4.3.1 Datasheet-Constrained Feedback Network

Because the AD8018 is a current-feedback amplifier, its feedback resistor is a stability-setting component rather than a purely gain-setting resistor. For a gain of +2, the datasheet recommends

$$R_F = 750 \, \Omega, \quad R_G = 750 \, \Omega \quad (29)$$

and these values were adopted as the fixed starting point for the output stage. This choice prevents the loop bandwidth from being increased artificially by reducing R_F into a region that could degrade phase margin or lead to instability.

In the present circuit, these correspond to

$$R_1 = 750 \, \Omega, \quad R_2 = 750 \, \Omega \quad (30)$$

for the AD8018 feedback network.

4.3.2 Feedback-Lead Compensation

Feedback-lead compensation was implemented by placing a capacitor C_f across the AD8018 feedback resistor. Using the design relation introduced in Section 2.4,

$$C_f = \frac{\sqrt{1+R_2/R_1}}{2\pi R_2 f_x} \quad (31)$$

and substituting $R_1 = R_2 = 750 \, \Omega$ gives

$$C_f = \frac{\sqrt{2}}{2\pi(750)f_x} \quad (32)$$

where f_x is the intended loop crossover frequency.

As an example, for a target crossover frequency of $f_x = 3 \text{ MHz}$,

$$C_f = \frac{1.414}{2\pi(750)(3 \times 10^6)} = 1.00 \times 10^{-10} \text{ F} \approx 100 \text{ pF} \quad (33)$$

Rather than repeating the same substitution for every trial crossover frequency, the corresponding capacitor estimates are summarized in Table 5.

Table 5: Feedback-lead capacitor estimates for candidate target crossover frequencies.

f_x	C_f
6 MHz	50 pF
5 MHz	60 pF
4 MHz	75 pF
3 MHz	100 pF
2 MHz	150 pF
1.5 MHz	200 pF
1 MHz	300 pF

These values provide a practical design range for simulation-based refinement. In particular, $C_f = 68$ pF, 100 pF, and 150 pF were identified as useful trial values spanning relatively light, moderate, and stronger feedback-lead compensation.

4.3.3 Input-Lag Network

The input-lag branch was introduced to improve phase margin by reshaping the inverse feedback factor near crossover. Following the design relation introduced in Section 2.4,

$$C_c = \frac{5}{\pi R_c f_x} \quad (34)$$

a practical initial value of

$$R_c = 10 \text{ k}\Omega \quad (35)$$

was selected so that the resulting C_c values fall in a practical picofarad range for the desired crossover frequencies.

As an example, for $f_x = 3$ MHz,

$$C_c = \frac{5}{\pi(10^4)(3 \times 10^6)} = 5.31 \times 10^{-11} \text{ F} \approx 53 \text{ pF} \quad (36)$$

The corresponding C_c values for several candidate crossover frequencies are summarized in Table 6.

Table 6: Input-lag capacitor estimates for $R_c = 10 \text{ k}\Omega$ and candidate target crossover frequencies.

f_x	C_c
6 MHz	26.5 pF
5 MHz	31.8 pF
4 MHz	39.8 pF
3 MHz	53.1 pF
2 MHz	79.6 pF
1.5 MHz	106 pF
1 MHz	159 pF

These estimates indicate that, for $R_c = 10 \text{ k}\Omega$, standard capacitor values such as 39 pF, 56 pF, 82 pF, and 100 pF provide a practical initial design set for the input-lag branch.

4.3.4 Initial Design Range for Simulation Refinement

Based on the above calculations, the following candidate values were selected as the initial design range for compensated voltage-mode loop-gain simulations:

$$C_f \in \{68 \text{ pF}, 100 \text{ pF}, 150 \text{ pF}\} \quad (37)$$

$$R_c = 10 \text{ k}\Omega \quad (38)$$

$$C_c \in \{39 \text{ pF}, 56 \text{ pF}, 82 \text{ pF}, 100 \text{ pF}\} \quad (39)$$

4.4 Compensated Voltage-Mode Loop-Gain Results

The compensated voltage-mode loop gain was evaluated across the same capacitive-load range. LTspice provides the circuit-level evidence, while MATLAB post-processing is used to extract readable stability metrics from the sweep.

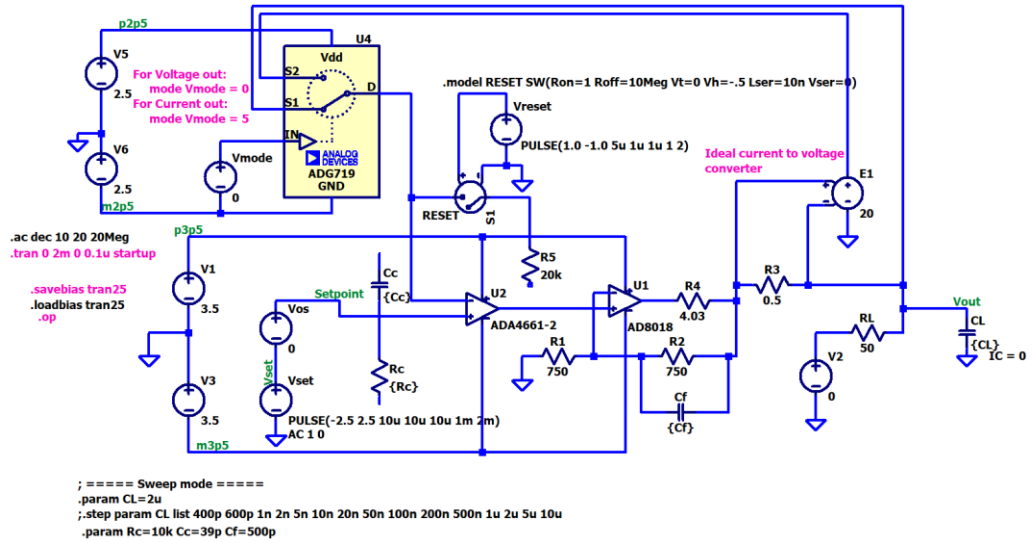


Figure 4.6: Final compensated voltage-mode circuit showing the adopted feedback-lead capacitor C_f and the input-lag branch, which was evaluated but not retained as the dominant stabilization mechanism.

4.5 Compensated Voltage-Mode Transient Results

The compensated voltage-mode configuration is shown in Figure 4.6. In this design, the input-lag branch was evaluated but did not contribute significantly to the overall stability improvement, whereas the feedback-lead capacitor across the AD8018 feedback path provided the dominant stabilization effect. Initial feedback-lead values based on the earlier design range improved the loop only partially, leaving many cases marginally stable. The capacitor was therefore tuned upward, and a value of $C_f = 500$ pF was selected as a practical compromise. At this value, the loop becomes stable for most of the required capacitive-load range while only four cases, 50 nF, 100 nF, 200 nF, and 500 nF, remain marginally stable. Increasing C_f further would likely move these cases into the fully stable region, but this was not pursued because their phase margins were already reasonably close to the target and further compensation would reduce crossover frequency and risk degrading settling performance.

The corresponding compensated loop-gain results are shown in Figure 4.7 from LTspice and in Figure 4.8 after MATLAB post-processing of the exported data. The extracted stability metrics in Figure 4.9 confirm that the compensated loop is now stable for eleven of the fifteen capacitive-load cases, with only the intermediate values of 50 nF, 100 nF, 200 nF, and 500 nF classified as marginally stable. Thus, compared with the uncompensated case, the compensated design shows a substantial improvement in voltage-mode robustness across the required capacitive-load range.

To assess the time-domain consequence of this compensation choice, representative transient cases were simulated and then processed in MATLAB. Figure 4.10 shows the compensated voltage-mode transient responses for 10 nF, 50 nF, 500 nF, 2 μ F, and 10 μ F, while the extracted transient metrics are summarized in Figure 4.11. These results show that the compensated design remains well controlled in the time domain, with moderate overshoot and finite settling on the falling transition. The retained marginal cases therefore represent a tradeoff between stronger stability margins and preserving faster dynamic response.

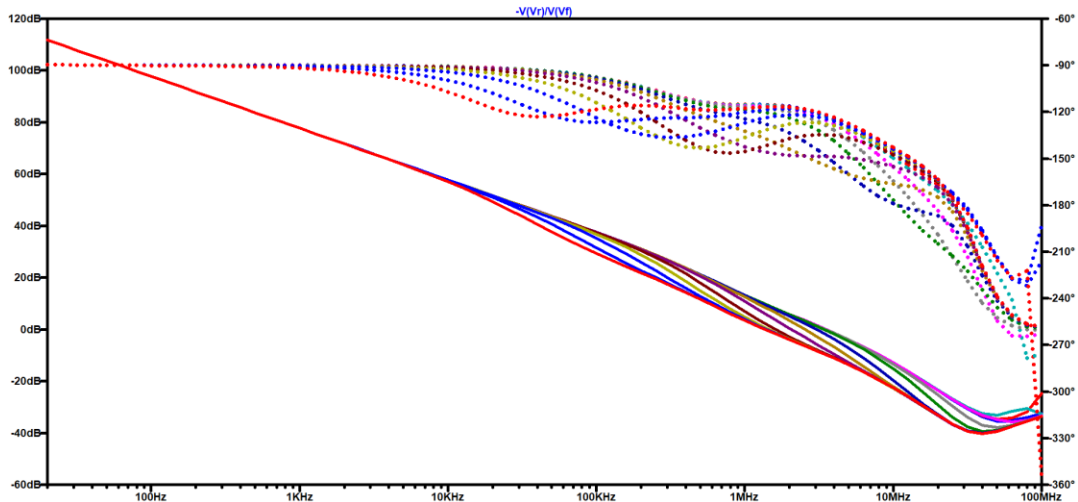


Figure 4.7: LTspice compensated voltage-mode loop-gain plot obtained using the Middlebrook voltage-injection method across the capacitive-load sweep.

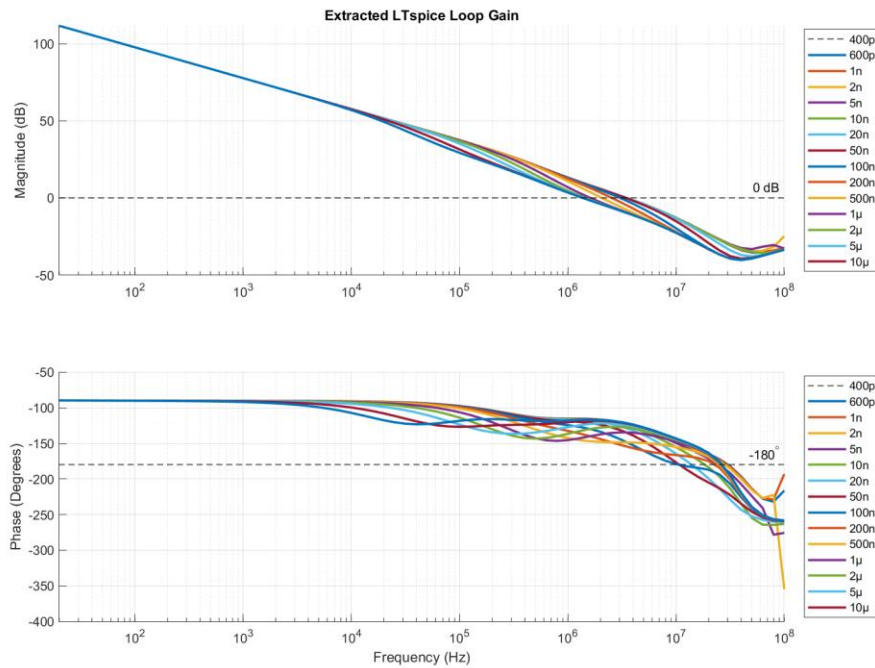


Figure 4.8: MATLAB-processed Bode plot of the compensated voltage-mode loop gain generated from the exported LTspice data.

	CL	f _{gc} (MHz)	f _{pc} (MHz)	GM (dB)	PM (deg)	Status
1	400p	3.5229	30.1677	30.0910	57.5430	Stable
2	600p	3.5232	29.4781	29.6340	57.4080	Stable
3	1n	3.5237	28.1216	28.7490	57.1370	Stable
4	2n	3.5245	25.0959	26.8480	56.4610	Stable
5	5n	3.5222	18.4704	22.2960	54.4550	Stable
6	10n	3.5032	13.9100	18.7540	51.2400	Stable
7	20n	3.4183	10.7449	16.4940	45.6370	Stable
8	50n	3.0376	10.6737	20.9910	35.9370	Marginally Stable
9	100n	2.5495	22.6768	35.2900	31.4240	Marginally Stable
10	200n	2.0632	25.2686	36.8220	31.9800	Marginally Stable
11	500n	1.6147	26.0662	37.1060	40.1190	Marginally Stable
12	1μ	1.4527	26.3137	37.1870	48.9560	Stable
13	2μ	1.3919	26.4340	37.2250	56.1080	Stable
14	5μ	1.3734	26.5042	37.2460	61.2770	Stable
15	10μ	1.3714	26.5260	37.2520	63.0780	Stable

Figure 4.9: Stability metrics extracted in MATLAB from the exported compensated voltage-mode loop-gain data, showing that only 50 nF, 100 nF, 200 nF, and 500 nF remain marginally stable.

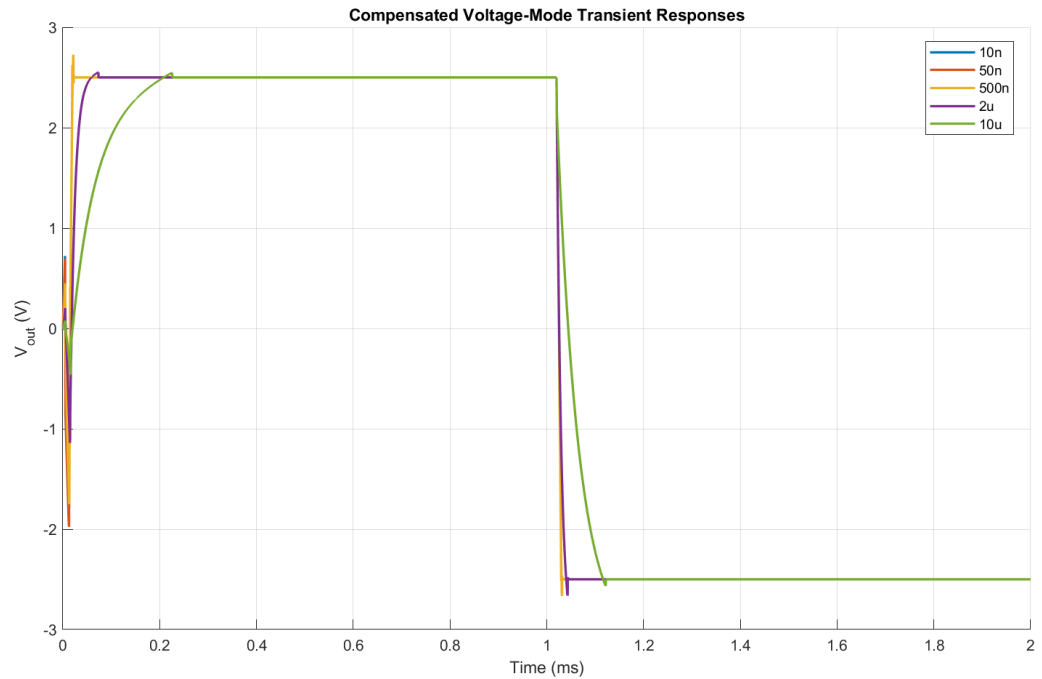


Figure 4.10: Compensated voltage-mode transient responses for representative capacitive loads used to assess time-domain damping and settling behavior.

	CL	OS rise (%)	US fall (%)	t _s rise (us)	t _s fall (us)
1	10n	4.7400	0.4700	NaN	7.7500
2	50n	4.8600	0.6600	NaN	9.6920
3	500n	9.0900	6.6100	NaN	11.0980
4	2u	2.0900	6.5200	NaN	22.6060
5	10u	1.8700	2.6700	NaN	101.1860

Figure 4.11: Transient metrics extracted in MATLAB for the representative compensated voltage-mode cases, including overshoot, undershoot, and settling time.

5 Current-Mode Design, Stability, and Compensation

5.1 Baseline Configuration and Verification

This section presents the baseline current-mode configuration used as the starting point for the current-loop stability study.

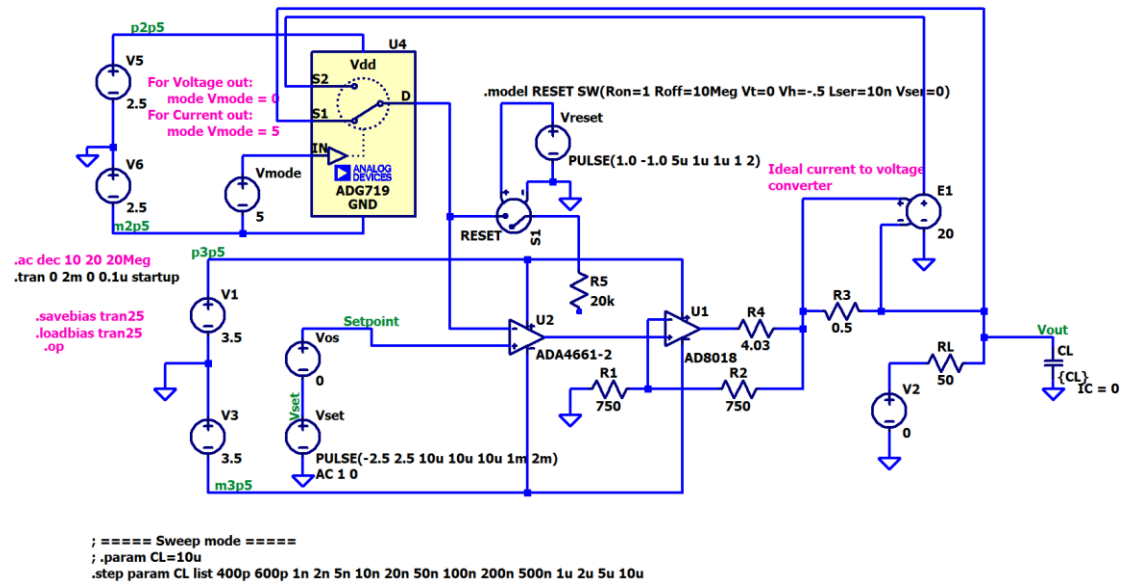


Figure 5.1: Baseline uncompensated circuit configured in current-output mode for initial current-mode verification and stability evaluation.

5.2 Uncompensated Stability Across Capacitive Load

The uncompensated current-mode configuration is shown in Figure 5.1. The time-domain response across the capacitive-load sweep is shown in Figure 5.2. Although the output current reaches the

intended steady-state levels for the smaller capacitive loads, the intermediate and larger load cases exhibit strong oscillatory behavior and pronounced current peaking, indicating that the uncompensated current loop is poorly damped.

The corresponding loop-gain behavior is shown in Figure 5.3 from LTspice and in Figure 5.4 after MATLAB post-processing of the exported data. Both figures show a much more severe stability degradation than in the uncompensated voltage-mode case. In particular, the gain crossover shifts upward sharply once the capacitance enters the intermediate and large-load region, while the phase margin collapses. The extracted stability metrics in Figure 5.5 confirm this trend. Only the smallest capacitive loads, 400 pF, 600 pF, 1 nF, and 2 nF, remain stable. All larger cases from 5 nF through 10 μ F are unstable, with negative gain margin and negative phase margin.

These results show that the uncompensated current-mode loop is substantially less robust than the uncompensated voltage-mode loop. In current mode, compensation is therefore not optional but necessary before the design can satisfy the required capacitive-load range.

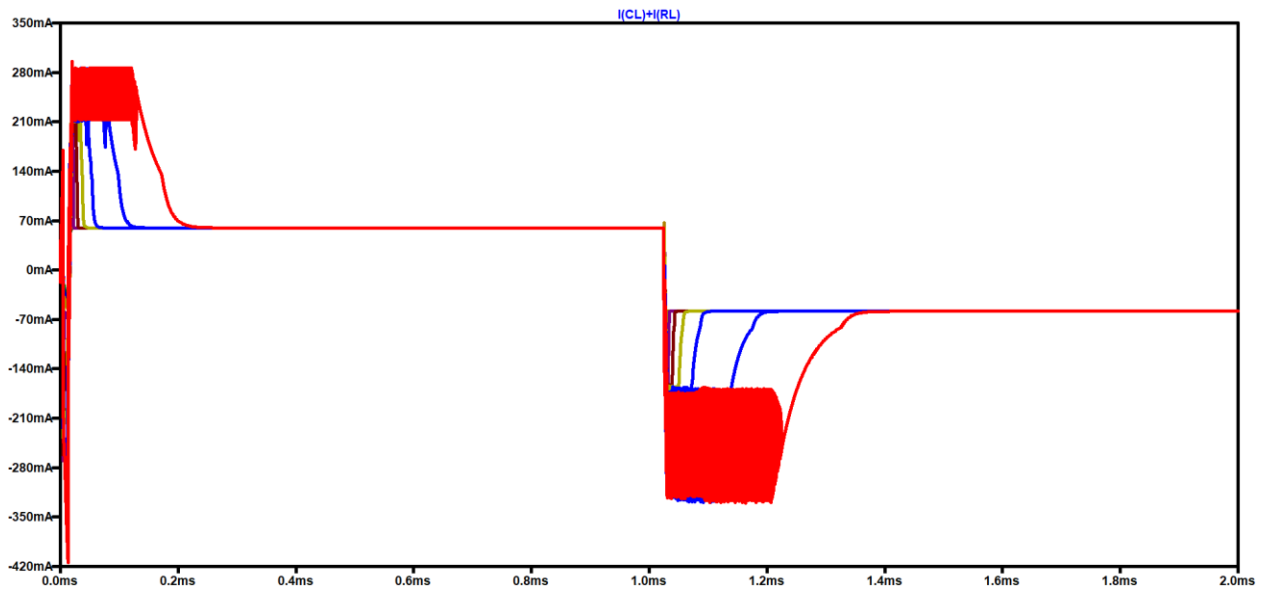


Figure 5.2: Uncompensated current-mode transient response across the capacitive-load sweep, showing strong oscillatory behavior and poor damping for intermediate and larger capacitive loads.

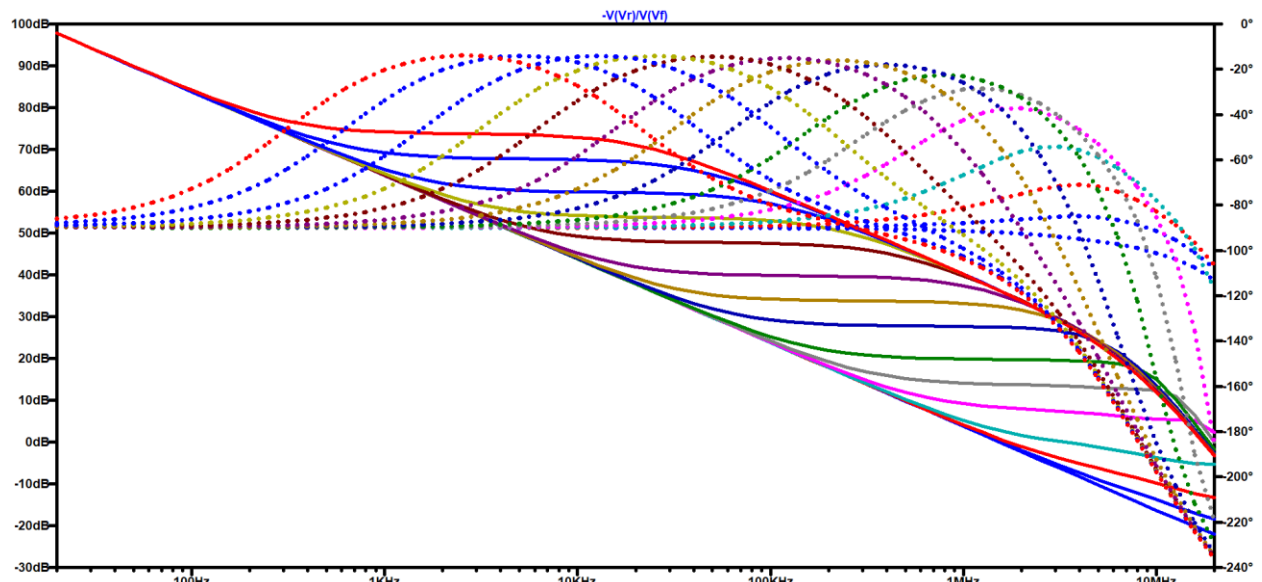


Figure 5.3: LTspice uncompensated current-mode loop-gain plot obtained using the Middlebrook voltage-injection method across the capacitive-load sweep.

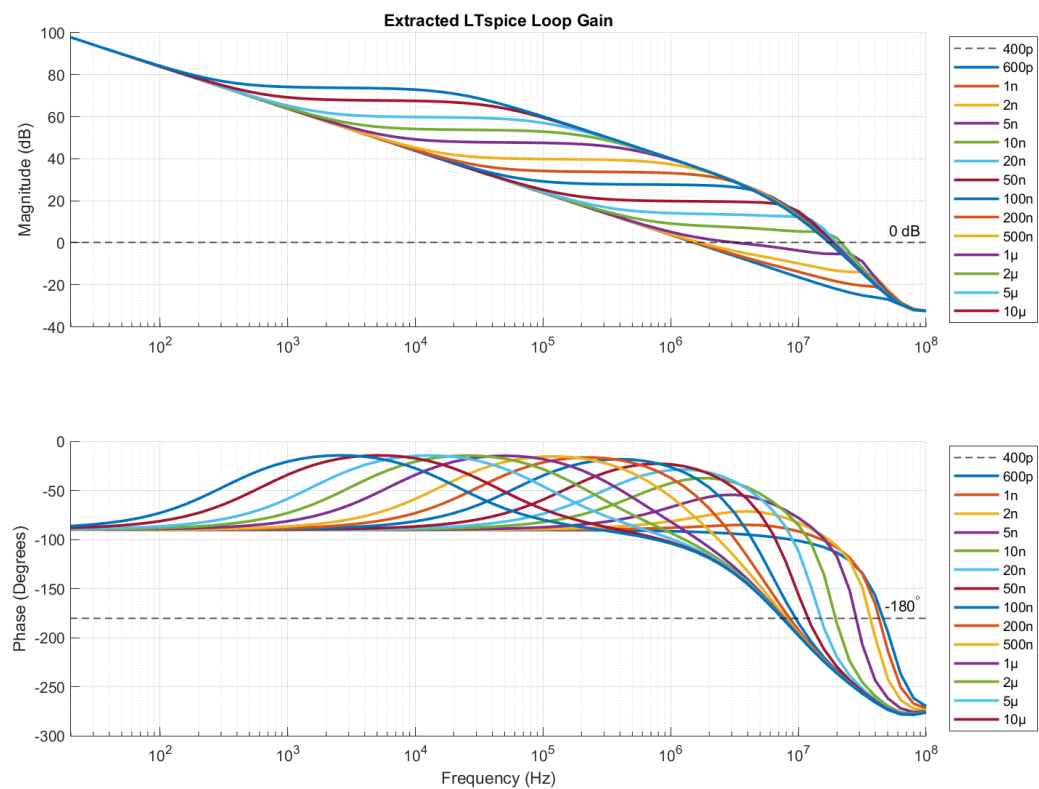


Figure 5.4: MATLAB-processed Bode plot of the uncompensated current-mode loop gain generated from the exported LTspice data.

	CL	f _{gc} (MHz)	f _{pc} (MHz)	GM (dB)	PM (deg)	Status
1	400p	1.5387	45.9421	26.5680	87.9570	Stable
2	600p	1.5761	42.7842	21.6700	93.1830	Stable
3	1n	1.7164	36.5229	15.1720	103.6130	Stable
4	2n	3.2988	28.3110	7.0620	125.5580	Stable
5	5n	21.5790	19.5044	-2.6020	-18.0870	Unstable
6	10n	19.9512	14.9938	-7.5790	-39.2490	Unstable
7	20n	18.7530	11.9014	-11.4200	-44.6180	Unstable
8	50n	17.9531	9.4927	-14.5460	-47.5280	Unstable
9	100n	17.6744	8.5341	-15.7910	-48.3770	Unstable
10	200n	17.5333	8.0020	-16.5350	-48.7770	Unstable
11	500n	17.4481	7.6729	-16.9300	-49.0080	Unstable
12	1μ	17.4197	7.5589	-17.0680	-49.0840	Unstable
13	2μ	17.4055	7.5010	-17.1390	-49.1220	Unstable
14	5μ	17.3969	7.4661	-17.1820	-49.1430	Unstable
15	10μ	17.3941	7.4546	-17.1960	-49.1500	Unstable

Figure 5.5: Stability metrics extracted in MATLAB from the exported uncompensated current-mode loop-gain data, showing stable behavior only for 400 pF, 600 pF, 1 nF, and 2 nF, with all larger capacitive loads unstable.

5.3 Current-Mode Compensation Strategy

The same compensation topology developed for voltage mode was used as the starting point for the current-mode design, since the feedback-lead capacitor across the AD8018 feedback path provided the dominant stabilization effect while the input-lag branch did not contribute significantly to the overall improvement. However, the uncompensated current-mode results showed that the current-sense path was more aggressive and less robust than in voltage mode. To reduce the effective gain of the sensing branch and improve loop stability, the sense resistor R_3 was reduced from 0.5Ω to 0.05Ω . This modification was adopted as the principal current-mode adjustment before re-evaluating the compensated loop gain and transient response.

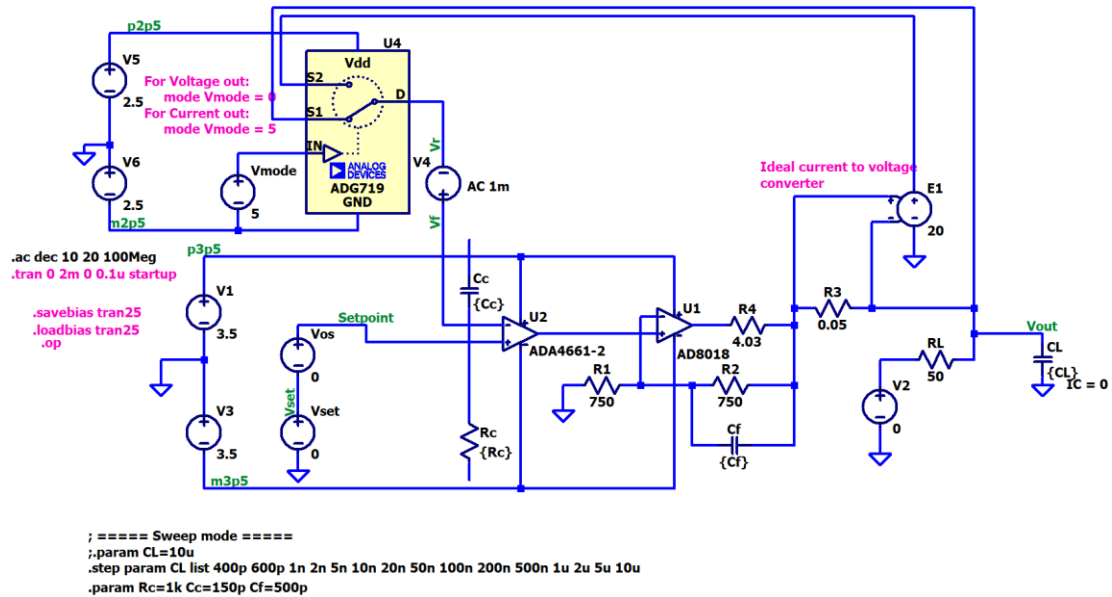


Figure 5.6: Final compensated current-mode circuit using the same compensation topology as voltage mode, with the current-sense resistor R_3 reduced from 0.5Ω to 0.05Ω to improve current-loop stability.

5.4 Compensated Current-Mode Loop-Gain Results

The compensated current-mode configuration is shown in Figure 5.6. The same compensation topology used in voltage mode was retained, with the main current-mode adjustment being the reduction of the sense resistor R_3 from $0.5\ \Omega$ to $0.05\ \Omega$ in order to reduce the effective gain of the sensing path and improve current-loop stability. The corresponding loop-gain results are shown in Figure 5.7 from LTspice and in Figure 5.8 after MATLAB post-processing of the exported data. The extracted stability metrics in Figure 5.9 show a clear improvement relative to the uncompensated current-mode case. The loop is stable from $400\ \text{pF}$ through $100\ \text{nF}$, while the larger capacitive loads $200\ \text{nF}$, $500\ \text{nF}$, $1\ \mu\text{F}$, $2\ \mu\text{F}$, $5\ \mu\text{F}$, and $10\ \mu\text{F}$ remain marginally stable. Thus, the compensation substantially increases the stable operating region, although the large-capacitance end of the range still exhibits limited phase margin.

The time-domain consequence of this compensated design is shown in Figure 5.10 for representative capacitive loads. The current responses remain bounded and settle to the commanded levels, but the larger capacitive cases exhibit substantial overshoot and undershoot, especially during the switching transitions. The extracted transient metrics in Figure 5.11 confirm this behavior: the falling-edge settling times remain finite, but the overshoot and undershoot grow rapidly with capacitance, and the rising-edge settling time is not obtained within the chosen 2% criterion for the representative cases processed in MATLAB. These results indicate that the compensated current-mode loop is significantly improved in stability, but its transient performance is still more challenging than in voltage mode, particularly for larger capacitive loads.

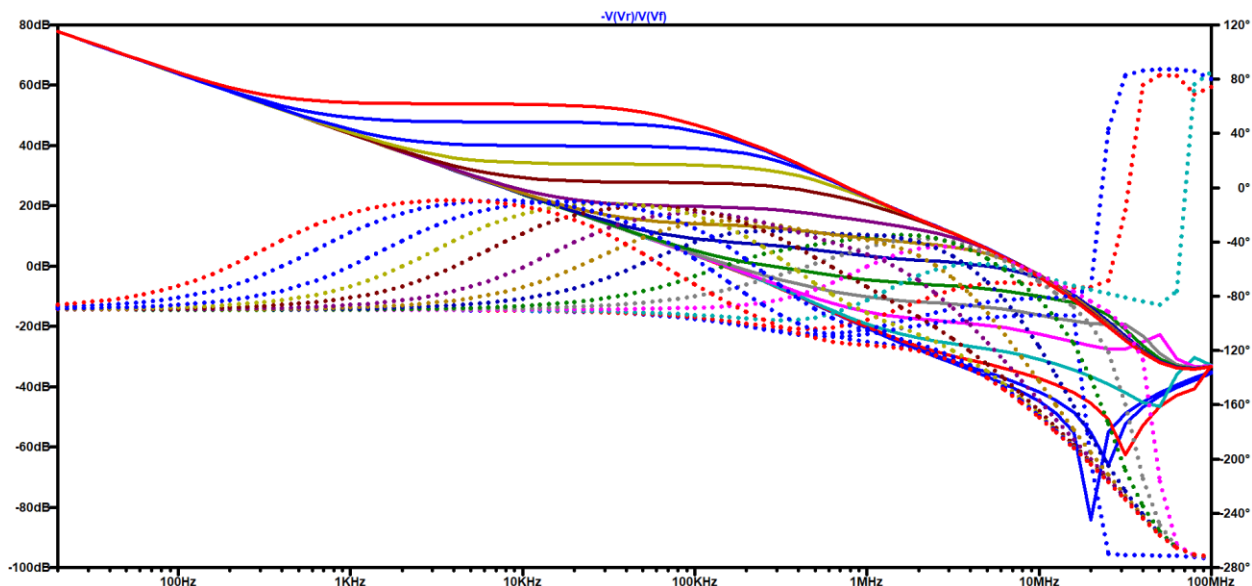


Figure 5.7: LTspice compensated current-mode loop-gain plot obtained using the Middlebrook voltage-injection method across the capacitive-load sweep.

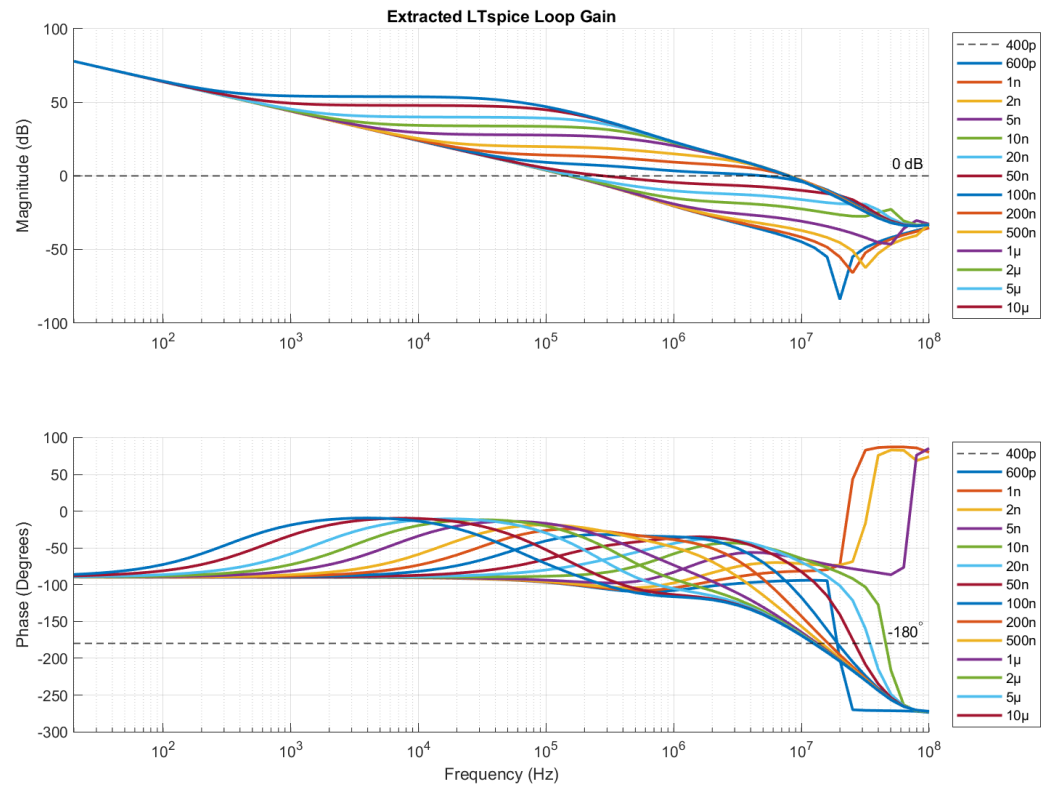


Figure 5.8: MATLAB-processed Bode plot of the compensated current-mode loop gain generated from the exported LTspice data.

	CL	f _{gc} (MHz)	f _{pc} (MHz)	GM (dB)	PM (deg)	Status
1	400p	0.1490	19.0630	78.083	80.443	Stable
2	600p	0.1490	Out of range	Out of range	80.979	Stable
3	1n	0.1491	Out of range	Out of range	82.050	Stable
4	2n	0.1496	Out of range	Out of range	84.717	Stable
5	5n	0.1530	45.7353	23.714	92.657	Stable
6	10n	0.1670	34.5061	20.727	105.931	Stable
7	20n	0.2752	26.1407	17.075	132.131	Stable
8	50n	4.9441	19.2320	12.965	110.994	Stable
9	100n	7.8875	16.2337	10.652	55.412	Stable
10	200n	8.0643	14.4852	9.172	36.762	Marginally Stable
11	500n	7.9425	13.3101	8.065	27.427	Marginally Stable
12	1μ	7.8667	12.8973	7.653	24.583	Marginally Stable
13	2μ	7.8235	12.6875	7.438	23.201	Marginally Stable
14	5μ	7.7959	12.5583	7.310	22.385	Marginally Stable
15	10μ	7.7865	12.5147	7.269	22.117	Marginally Stable

Figure 5.9: Stability metrics extracted in MATLAB from the exported compensated current-mode loop-gain data, showing stable behavior up to 100 nF and marginally stable behavior for the larger capacitive loads.

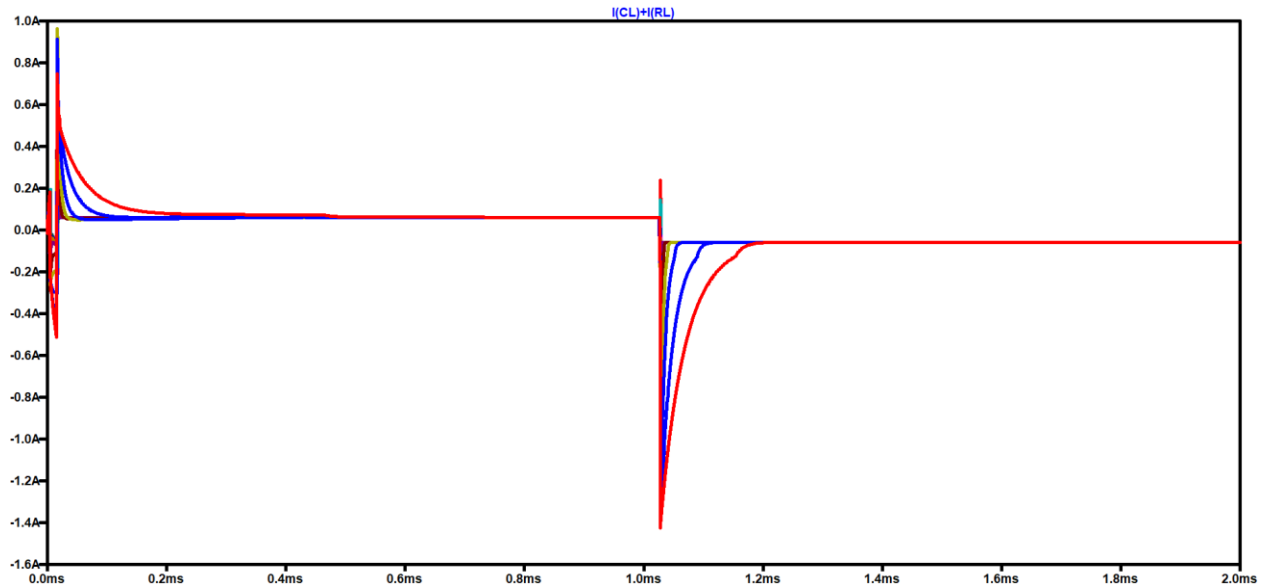


Figure 5.10: Compensated current-mode transient responses for representative capacitive loads used to assess time-domain damping and settling behavior.

	CL	OS rise (%)	US fall (%)	t _s rise (us)	t _s fall (us)
1	10n	49.8500	62.2300	N/A	1.907
2	50n	266.9400	279.2900	N/A	2.323
3	500n	1.4090e+03	1.9945e+03	N/A	9.616
4	2u	1.4489e+03	2.2387e+03	N/A	37.073
5	10u	1.0085e+03	2.3244e+03	N/A	180.103

Figure 5.11: Transient metrics extracted in MATLAB for the representative compensated current-mode cases, including overshoot, undershoot, and settling time.

6 ECSE 6050 Extension: Inductive-Load Results

6.1 Load Model and Rationale

For the required ECSE 6050 extension, the same load model used in the capacitive-load study was retained, except that the output capacitor was replaced by an inductor. The inductance L was then parameterized and swept up to 100 mH to evaluate the effect of inductive loading on loop stability and transient response.

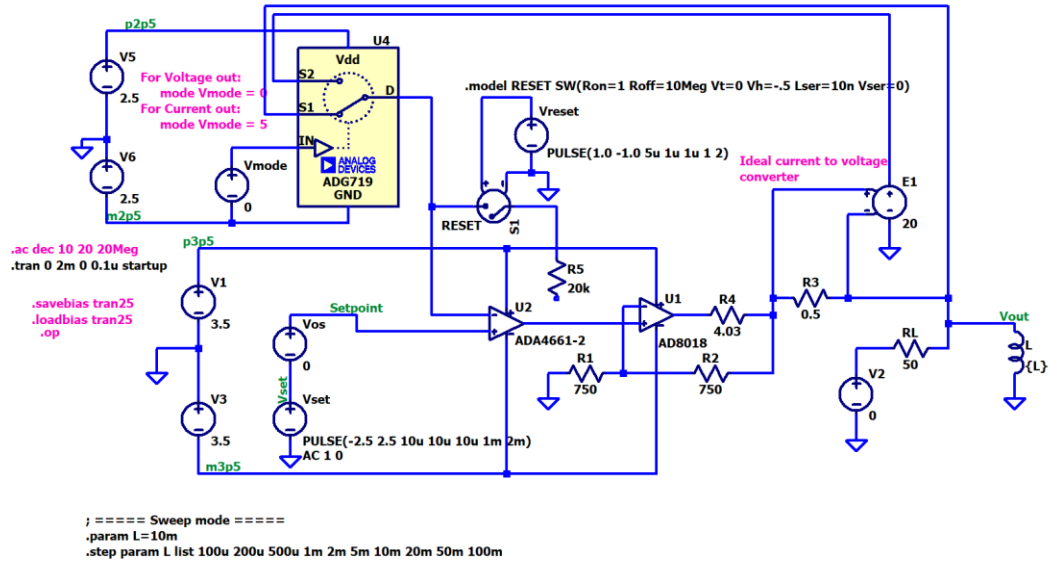


Figure 6.1: Inductive-load model used for the required ECSE 6050 extension, obtained by replacing the output capacitor in the previous load model with a parameterized inductor L swept up to 100 mH.

6.2 Current Mode with Inductive Loads

The current-mode RL-load results are shown in Figures 6.2 to 6.7. The same compensated current-mode circuit used in the RC study was retained, with the AD8018 feedback network set to the datasheet-recommended starting values of 750 Ω . The transient response across the inductive-load sweep is shown in Figure 6.2. The output current remains bounded for all inductance values, although the smaller inductances show larger initial peaking, while the larger inductances produce slower current ramps and longer decay intervals.

The corresponding loop-gain behavior is shown in Figure 6.3 from LTspice and in Figure 6.4 after MATLAB post-processing of the exported data. In contrast to the RC current-mode case, the RL current-mode loop remains well behaved across the full inductive-load range. The extracted stability metrics in Figure 6.5 confirm that all tested inductance values from 100 μH to 100 mH are stable, with consistently large gain margin and phase margin. Thus, under RL loading, the compensated current-mode design satisfies the stability requirement over the entire swept range.

The time-domain response used for transient assessment is shown in Figure 6.6, and the extracted transient metrics are summarized in Figure 6.7. These results show that the falling-edge settling time increases significantly as the inductance increases, which is expected from the slower current decay imposed by larger inductive loads. Overshoot and undershoot are largest at the smaller inductances, where the current changes most rapidly, while the larger inductances show slower but more gradual responses. Overall, the RL current-mode case is stable across the full range, but with the expected tradeoff that larger inductance leads to slower transient settling.

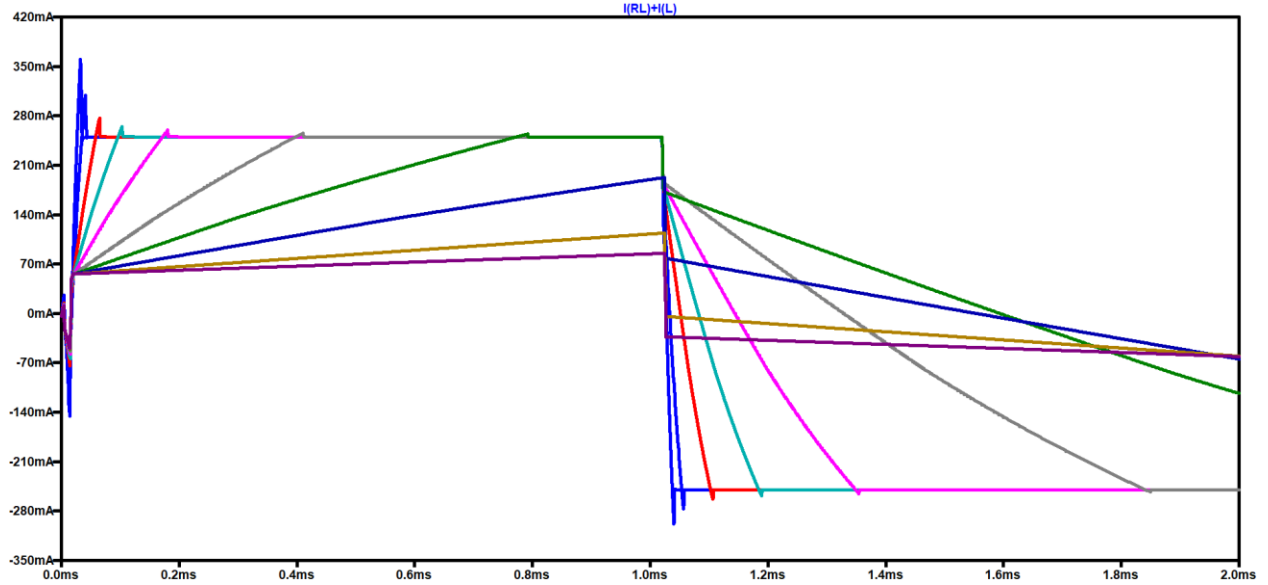


Figure 6.2: Compensated current-mode RL-load transient response across the swept inductive-load range, showing bounded current behavior for all tested inductances.

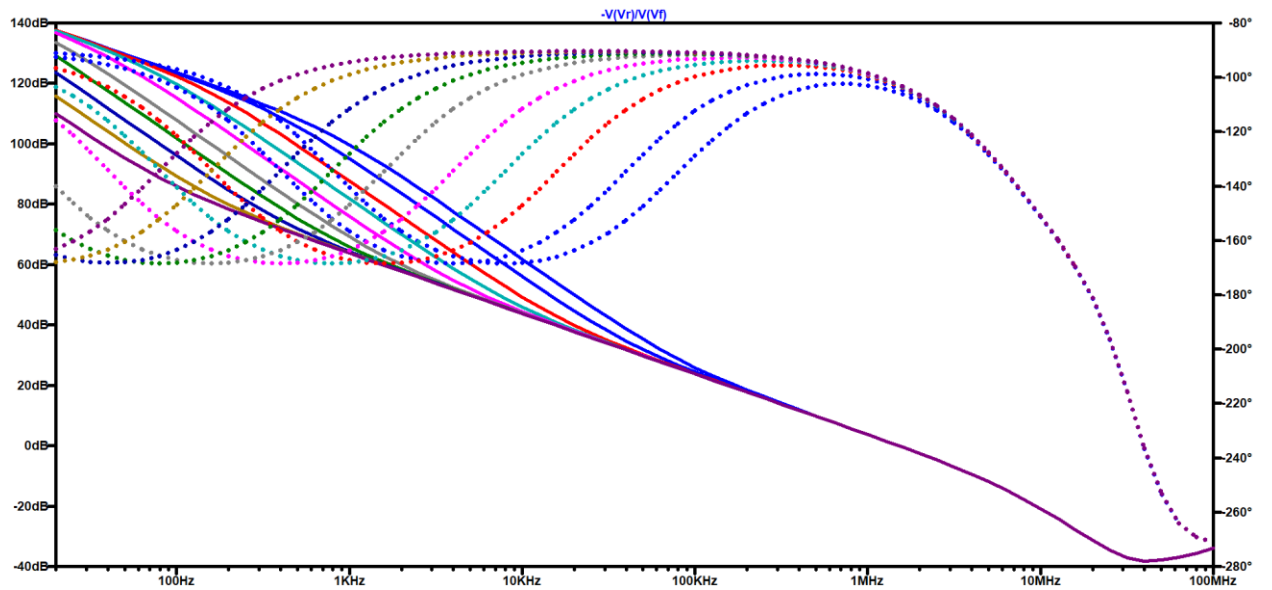


Figure 6.3: LTspice compensated current-mode RL-load loop-gain plot obtained using the Middlebrook voltage-injection method across the inductive-load sweep.

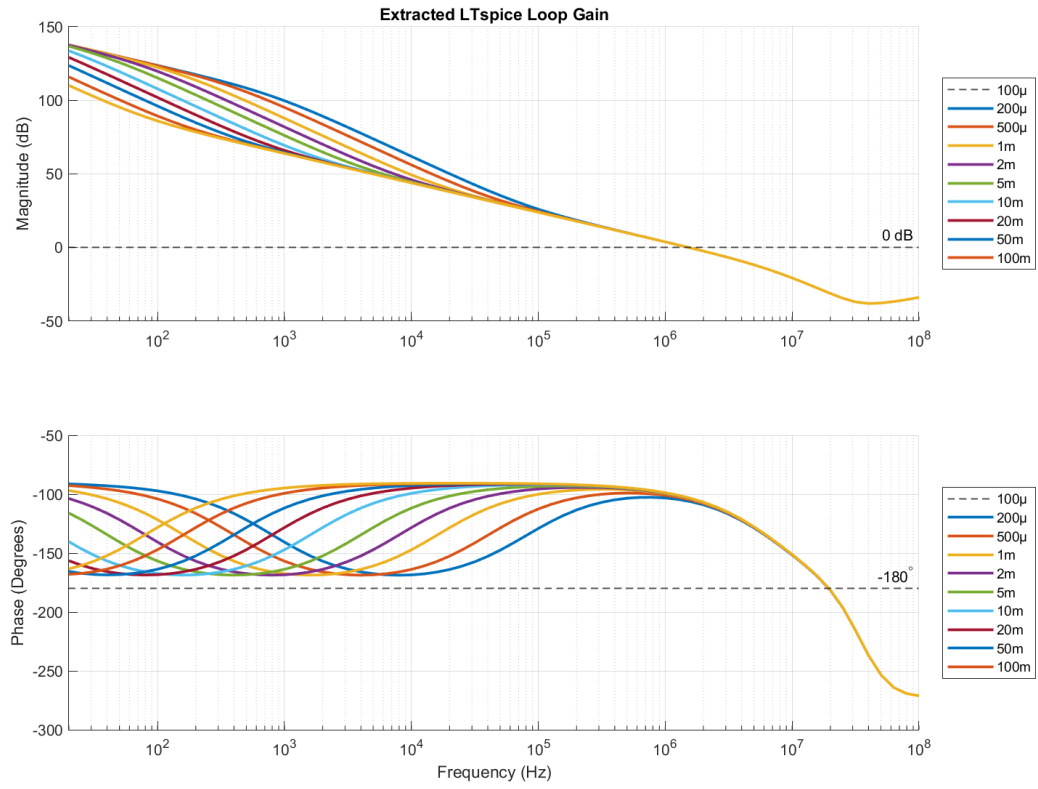


Figure 6.4: MATLAB-processed Bode plot of the compensated current-mode RL-load loop gain generated from the exported LTspice data.

	L	f _{gc} (MHz)	f _{pc} (MHz)	GM (dB)	PM (deg)	Status
1	100μ	1.5100	19.4310	30.926	74.323	Stable
2	200μ	1.5088	19.4761	30.962	75.831	Stable
3	500μ	1.5086	19.5031	30.984	76.735	Stable
4	1m	1.5086	19.5121	30.991	77.036	Stable
5	2m	1.5086	19.5166	30.995	77.186	Stable
6	5m	1.5087	19.5193	30.997	77.276	Stable
7	10m	1.5087	19.5202	30.998	77.306	Stable
8	20m	1.5087	19.5207	30.998	77.321	Stable
9	50m	1.5087	19.5209	30.998	77.330	Stable
10	100m	1.5087	19.5210	30.998	77.333	Stable

Figure 6.5: Stability metrics extracted in MATLAB from the exported compensated current-mode RL-load loop-gain data, showing stable behavior across the full inductive-load sweep.

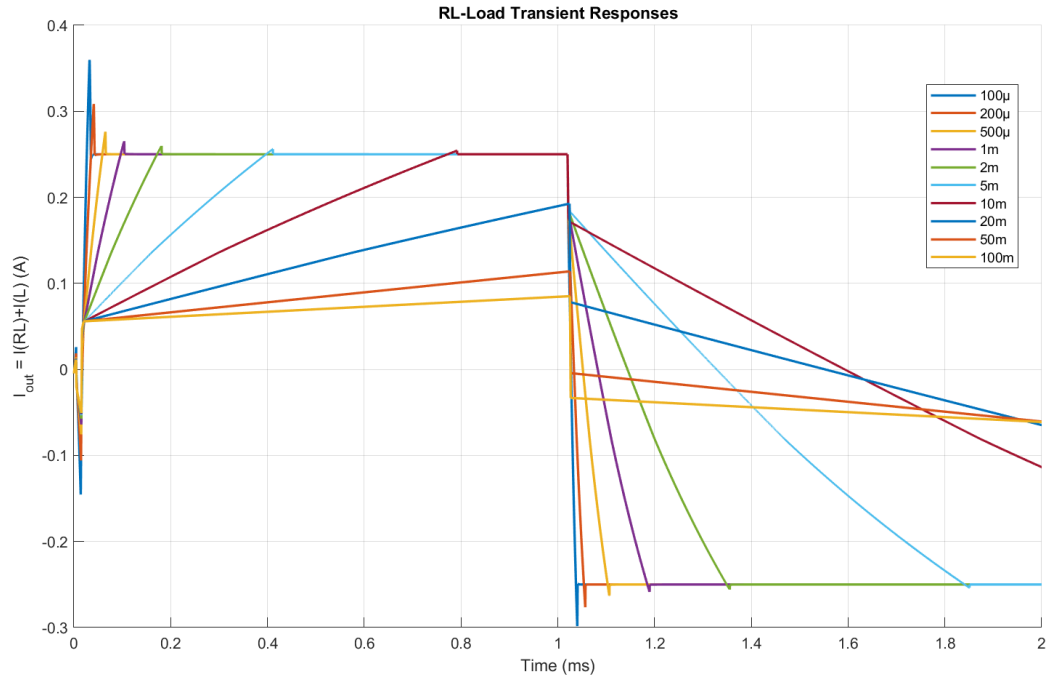


Figure 6.6: MATLAB-processed transient responses of the compensated current-mode RL-load case for the swept inductive-load range.

	L	OS rise (%)	US fall (%)	t _s rise (us)	t _s fall (us)
1	100μ	43.9000	19.3800	N/A	16.590
2	200μ	23.4000	10.6000	N/A	35.720
3	500μ	10.5200	5.2100	N/A	85.431
4	1m	6.1300	3.5000	N/A	168.391
5	2m	3.9200	2.4000	N/A	334.032
6	5m	8.8300	17.5100	N/A	N/A
7	10m	36.8900	116.5700	1005.702	N/A
8	20m	54.8400	100.0900	N/A	N/A
9	50m	36.0900	27.0700	N/A	N/A
10	100m	21.8000	11.7800	N/A	N/A

Figure 6.7: Transient metrics extracted in MATLAB for the compensated current-mode RL-load case, including overshoot, undershoot, and settling time across the inductive-load sweep.

6.3 Voltage Mode with Inductive Loads

The voltage-mode RL-load results are shown in Figures 6.8 to 6.12. The transient response in Figure 6.8 indicates that the loop remains bounded across the swept inductive-load range. For the smaller inductances, the response shows noticeable oscillatory behavior immediately after the switching transitions, but these oscillations are attenuated and decay well before the midpoint of the cycle. The Middlebrook voltage-injection result in Figure 6.9 and the combined voltage- and

current-injection comparison in Figure 6.10 do not provide a practical unity-gain crossover within the simulated frequency range, so a standard crossover-based phase-margin and gain-margin extraction was not used for this case. Accordingly, the RL voltage-mode assessment was based primarily on the observed transient behavior together with the qualitative loop-shape trends.

The MATLAB-processed transient responses in Figure 6.11 provide clearer labeling of the inductive-load sweep and show that the time-domain behavior depends strongly on inductance. The smaller inductances exhibit brief oscillatory transients that decay rapidly, whereas the larger inductances show much slower recovery after the switching transitions. The extracted transient metrics in Figure 6.12 confirm this trend. In particular, the smaller inductances show very large overshoot and undershoot, while the larger inductances exhibit reduced peaking but significantly slower response. For several of the lower-inductance cases, a 2% settling time is not obtained within the simulation window, indicating that although the response is bounded, it is not tightly damped under the chosen criterion. Overall, the voltage-mode RL response remains bounded over the tested inductive-load range, but its transient performance is clearly less well damped at the low-inductance end and progressively slower at the high-inductance end.

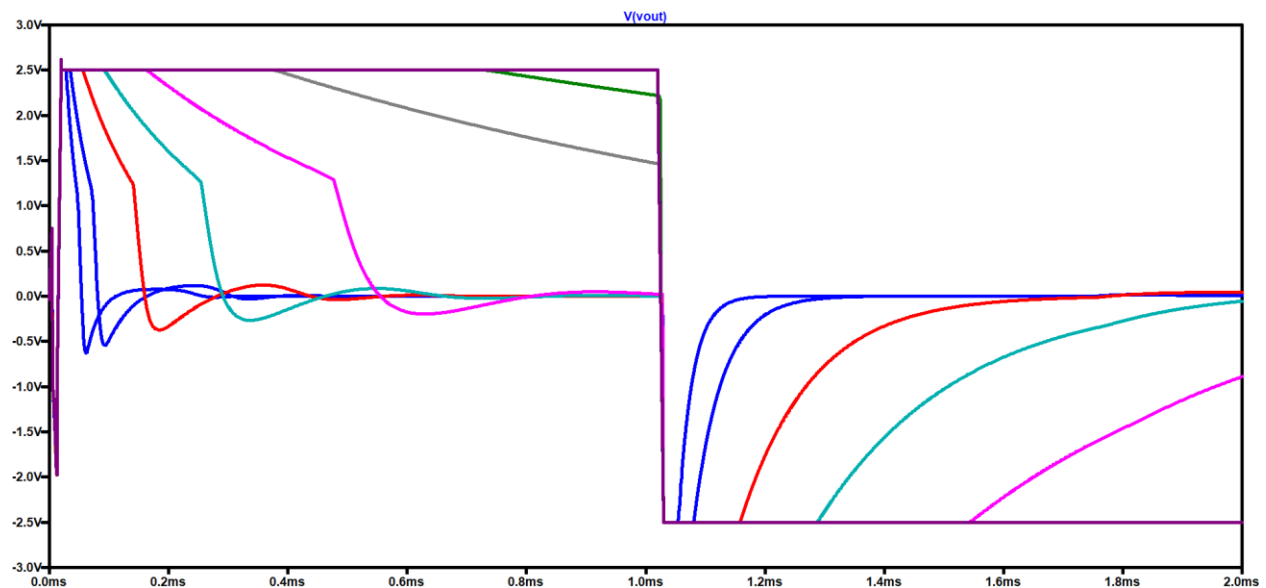


Figure 6.8: Voltage-mode RL-load transient response across the swept inductive-load range, showing bounded behavior with decaying oscillations for the smaller inductances.

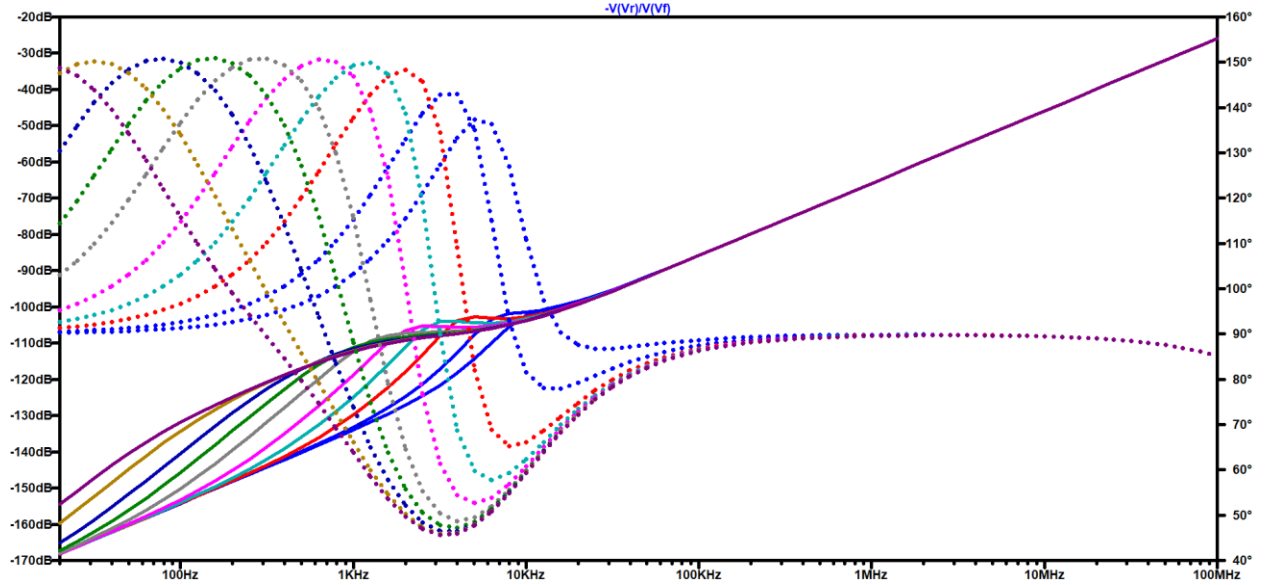


Figure 6.9: LTspice voltage-injection Middlebrook plot for the voltage-mode RL-load case across the inductive-load sweep. No practical unity-gain crossover is observed within the simulated frequency range.

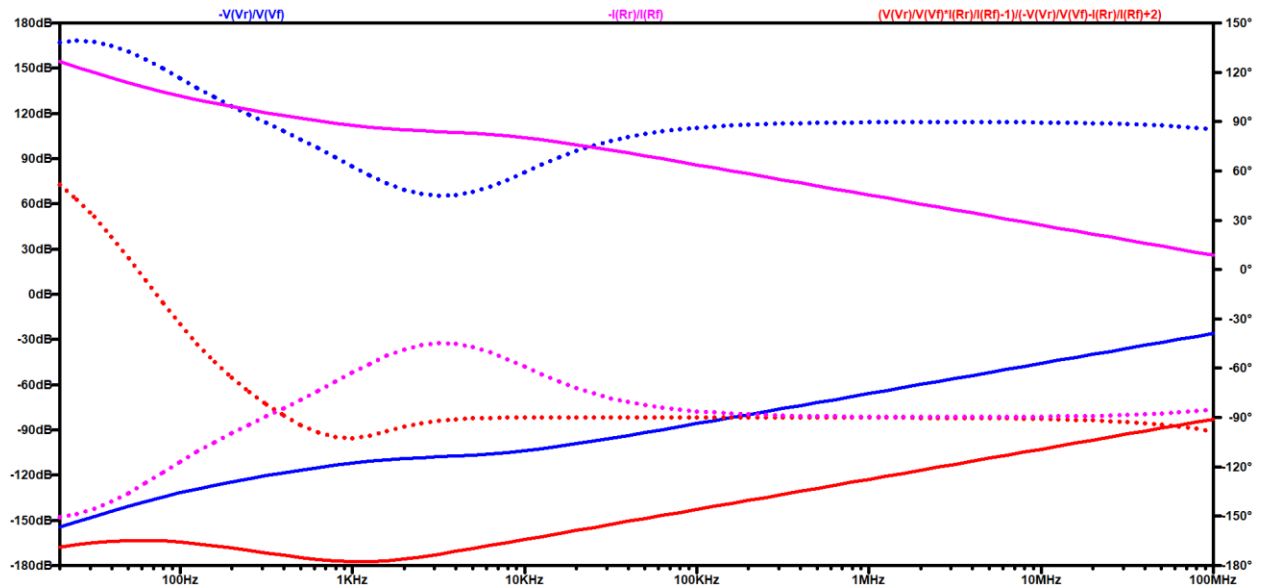


Figure 6.10: Comparison of voltage injection, current injection, and assembled loop gain for the voltage-mode RL-load case. Although the loop-shape trends can be compared qualitatively, no practical crossover-based margin extraction was used in this case.

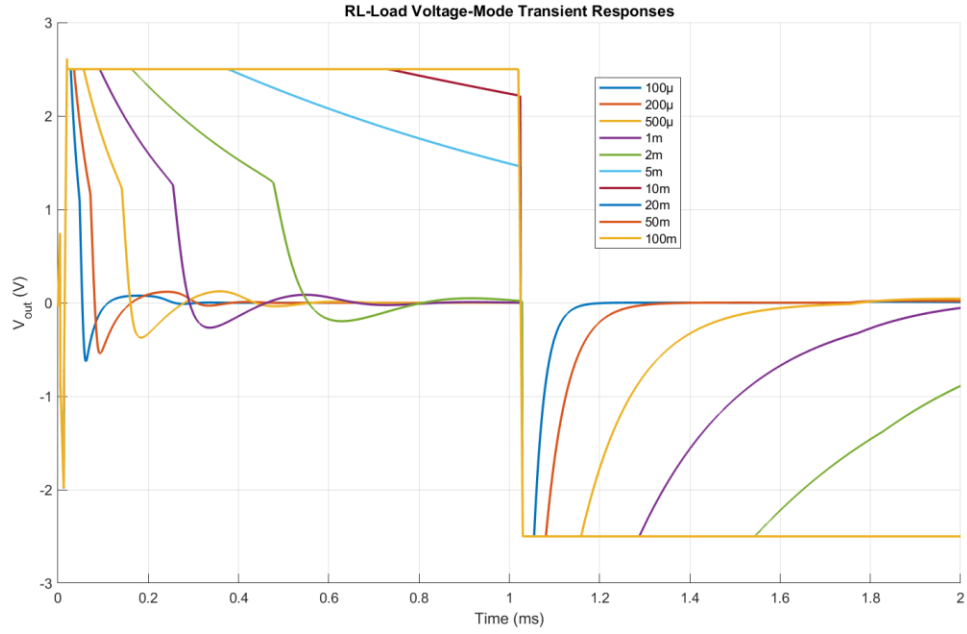


Figure 6.11: MATLAB-processed voltage-mode RL-load transient responses for the swept inductive-load range, showing the dependence of oscillation and recovery behavior on inductance.

	L	OS rise (%)	US fall (%)	t _s rise (us)	t _s fall (us)
1	100μ	6.2953e+04	4.3392e+04	N/A	N/A
2	200μ	1.7985e+04	2.2753e+04	N/A	N/A
3	500μ	7.9341e+04	3.4028e+05	N/A	N/A
4	1m	1.8725e+03	653.9100	N/A	N/A
5	2m	214.5300	60.1400	N/A	N/A
6	5m	17.5500	0.1400	N/A	4.767
7	10m	4.7200	0.1700	N/A	5.392
8	20m	4.5600	0.1500	N/A	7.791
9	50m	4.5600	0.1700	N/A	7.727
10	100m	4.5600	0.1800	N/A	7.793

Figure 6.12: Transient metrics extracted in MATLAB for the voltage-mode RL-load case, including overshoot, undershoot, and settling time across the inductive-load sweep

7 Comparison and Design Trade-Offs

7.1 Performance Summary

The final results show that the two operating modes do not present the same stability challenge. Under capacitive loading, voltage mode is more robust than current mode. After compensation, voltage mode became stable across most of the required capacitor range, with only a few intermediate cases remaining marginally stable. Current mode improved substantially, but the larger capacitive loads still produced reduced margins and poorer transient behavior.

Under inductive loading, the current-mode RL case behaved more favorably. It remained stable

across the full inductive sweep with strong margins, although settling became slower as inductance increased. The voltage-mode RL case remained bounded, but practical gain and phase margins could not be extracted from the loop-gain plots within the simulated frequency range, so its assessment relied mainly on transient behavior.

7.2 Discussion of Trade-Offs

The main tradeoff in this design was between stability and speed. Increasing compensation improved robustness, especially under capacitive loading, but reduced crossover frequency and slowed the response. This was most evident in voltage mode, where stronger feedback-lead compensation improved stability but reduced dynamic speed.

A second tradeoff appeared between frequency-domain stability and transient quality. In current mode, even when the loop became stable or marginally stable, several of the larger capacitive loads still showed significant overshoot and undershoot. Thus, acceptable margins did not always guarantee well-damped transient behavior.

The most difficult case overall was current mode with capacitive loading. By contrast, current mode under RL loading behaved much better and remained stable over the full sweep. Voltage mode under RL loading remained bounded, but its behavior had to be judged mainly from transient evidence rather than practical crossover-based margins.

8 Alternative IC Options

To ensure stable and accurate operation of the SMU, alternate ICs were considered based on key parameters such as offset voltage, input bias current, bandwidth, slew rate, output-current capability, and supply compatibility.

For the error-amplifier stage, the main objective was to identify precision amplifiers with low offset voltage and low input bias current so that measurement accuracy would be preserved. Table 7 summarizes few candidate options relative to the ADA4661-2.

Table 7: Comparing Error Amplifier Chip Options

Chip	Features
ADA4661-2	Low offset voltage: 150 μV typ; Low input bias current: 15 pA max; Low-power precision amplifier: $\sim 630 \mu\text{A}$ /amplifier typ; Bandwidth: 2.1 MHz typ
ADA4666-2	Low-power precision amplifier in the same general device family; Suitable as an alternate low-power precision option
LT6016	Low offset voltage: $\pm 50 \mu\text{V}$ max; Bandwidth: 3.2 MHz typ; Good precision-stage alternate with somewhat higher speed

For the output stage, the emphasis was on bandwidth, slew rate, and output-current capability, since this stage must drive fast-changing signals and reactive loads. Table 8 compares representative options relative to the AD8018.

Table 8: Comparing Output Stage Amplifier Chip Options

Chip	Features
AD8018	Bandwidth: 130 MHz typ; Slew rate: 300 V/ μ s typ; Current-feedback amplifier; Strong output-drive capability; Selected baseline
AD8397	Bandwidth: 69 MHz typ; Slew rate: 53 V/ μ s typ; High output current, but voltage-feedback architecture would require compensation redesign
LT1210	Bandwidth: 35 MHz typ; Slew rate: 900 V/ μ s typ; Very high output-current capability, but not compatible with the present low-voltage supply requirement

Overall, the LT6016 was identified as a practical alternate for the precision stage, while the AD8018 remained the most suitable output-stage choice because it best matched both the performance targets and the compensation framework used in this work.

9 Conclusion

This project designed and evaluated a configurable voltage/current source stage for an SMU using a composite amplifier architecture. The ADA4661-2 provided the precision front end, while the AD8018 provided the high-speed output stage. The circuit was verified in both voltage mode and current mode, and its stability was studied using Middlebrook loop-gain analysis together with transient simulation.

The uncompensated RC results showed that compensation was essential. Feedback-lead compensation around the AD8018 provided the main stabilization effect, while the input-lag branch had limited impact in the final design. In current mode, the sense resistor was reduced from 0.5 Ω to 0.05 Ω to improve loop stability. After compensation, voltage mode became stable over most of the required capacitive-load range, and current mode improved substantially, though the larger capacitive loads remained the most challenging cases.

For the ECSE 6050 extension, the design was also tested with inductive loads up to 100 mH. The compensated current-mode RL case remained stable across the full range, while the voltage-mode RL case remained bounded in the time domain even though practical crossover-based margin extraction was not available. Overall, the design met the main project objectives, with the primary limitations being reduced damping or slower settling in the most demanding reactive-load cases.

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